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Criteria/Weight	Supervisor's comments
Executive Summary (5%)	Mark:
· Standalone Clarity: Can the reader understand the business problem, the technical approach, and the final recommendation without reading the rest of the paper?	
· Value Proposition: Does it explicitly state the value of the research, the "so what" of the analysis? Is the value proposition specifically tailored to the problem at hand?	
· Is it concise (typically about 2 pages) while covering all major sections of the dissertation?	
Problem Framing & Scope (10%)	Mark:
· Specificity and relevance: Does the student successfully convert sometimes vague/ill-defined business issues/problems (e.g., 'the company is losing sales ') into a business analytics problem (e.g., "a supervised classification model to predict churn with a focus on recall")?	
· Hypotheses: Are there clear, testable hypotheses or research questions grounded in existing methodology and clearly linked to the project scope?	
· Success Metrics: Has the student defined what 'success' looks like technically (e.g., F1-score, a model with strong fit metrics) and have these metrics been linked to relevant organisational goals (e.g., cost savings, efficiencies)?	
Literature Review (10%)	Mark:
· Critical Synthesis: Does the student compare different models/theories, or is the review just a summary of vaguely similar applications?	
· Methodological Justification: Does the review lead logically to the chosen methodology? (e.g., 'X method was used in similar industry cases, but Y is better for this specific situation').	

Criteria/Weight	Supervisor's comments
· Breadth: Does it cover both academic theory and current industry/practitioner benchmarks?	
Methodology (20%)	Mark:
· Selection Logic: Is there a robust justification for the chosen algorithms/tools over alternatives?	
· Data Rigor: Is there a clear audit trail of data cleaning, handling of missing values, and feature engineering?	
· EDA (Exploratory Data Analysis): Is there a suitable and clear data summary section? Are distributions, correlations, and anomalies identified before modelling? Are the visualisations relevant and high quality?	
· Ethics & Governance (where relevant): Does the student address data privacy (GDPR), potential algorithmic bias, or ethical implications of their model's deployment?	

Results & Discussion (25%)	Mark:
· Graphical presentation: Are visualizations clean, well-labeled, insightful or just the default/unformatted software output?	
· Translation of Stats: Does the student clearly translate the technical results (e.g., a p-value or a coefficient) and describe their impact?	
· Actionable Recommendations: Are the suggestions specific? (e.g., "Target customers in the 18-24 bracket with a 10% discount" rather than "Improve marketing").	
· Feasibility: Does the discussion acknowledge the costs, time and general feasibility of implementing the suggested solution?	
Insight, Critical Evaluation & GenAI (20%)	Mark:
· Ownership: Does the student demonstrate they truly understand the underlying math/logic, or are they just "running code"?	
· Critical Reflection: Does the student identify the limitations of their work? Do they explain where the model might fail or where the data was insufficient?	

Results & Discussion (25%)	Mark:
· Narrative Flow: Does the student manage to tell a compelling story through data and analysis? Is there a clear flow from structuring the business problem through the analysis to the final conclusion?	
· GenAI audit and attribution: Is there a clear log of how GenAI was used? Does the student evaluate the AI's output for accuracy or bias?	
Complexity & Progress (10%)	Mark:
· Problem-Solving: Did the student overcome significant technical hurdles (e.g., communication failures, messy unstructured data, compute limitations)?	
· Technical Ambition: Is the project's scope appropriate for an MSc level? Does it move beyond basic "out-of-the-box" tutorials?	
· Persistence: Where the initial approach failed (which is not uncommon in practice), did the student document a logical pivot or an iterative improvement process?	

General marking guidelines	General marking guidelines	General marking guidelines	General marking guidelines
80 +	Outstanding work of publishable standard.	50 - 59	Good work which only covers a basic analysis. Some problems but no major omissions.
70 - 80	Excellent work showing mastery of the subject matter & excellent analytical skills.	40 - 49	Inadequate work. Not sufficiently analytical. Some major omissions.
60 - 69	Very good work. Interesting analysis with original insights. Some minor errors.	0 - 39	Work is seriously flawed. Lack of clarity & argumentation. Too descriptive.

FINAL MARK: _____

Operationalising ESG Reporting: The Design, Deployment and Evaluation of an Automated Platform for Consortium-Level Sustainability Data

Executive Summary

Environmental, social and governance reporting has become a regulatory obligation and an investor expectation, yet the processes that produce it remain substantially manual. Within consortia such as the World Business Council for Sustainable Development Tire Industry Project, member companies submit data through spreadsheets that analysts then consolidate, convert, validate and benchmark by hand, a process that is slow, error prone and difficult to govern. Two distinct bodies of research bear on this problem. One establishes that sustainability data is fragmented across incompatible units and reporting formats, and that reconciling it by hand is laborious. The other establishes that automation can improve business processes where the underlying work is redesigned rather than merely digitised. What neither body of work provides is a measured account of what actually changes when a real reporting workflow is automated and compared directly against the manual process it replaces. This dissertation addresses that gap.

The study adopts a Design Science Research strategy, developing a functioning web based reporting platform for the consortium, deployed on Microsoft Azure, and evaluating it against the manual process across four dimensions drawn from the literature, namely efficiency, data quality, productivity and credibility. In keeping with the iterative character of Design Science, the platform was built across successive cycles, from an initial build on synthetic data, through refinement of the schema and the unit conversion logic, to migration onto real client data and deployment. The final iteration is the one evaluated here. Evaluation combines a controlled accuracy test against a synthetic dataset whose correct values are known in advance with a parallel run in which three member companies completed a full reporting cycle under both the manual and the automated method.

Three findings result. The first concerns accuracy. Across 3,910 unit conversions the platform reproduced the correct value in every instance, and its validation rules surfaced 97 anomalies that manual inspection would otherwise have had to detect unaided. The second concerns efficiency. In the parallel run a manual cycle of roughly 18 to 21 hours for each company fell to 7 to 8 hours on the platform, a reduction of approximately two thirds, concentrated in the collection and analysis stages while the judgement dependent stages of validation and verification remained with the analyst. The third concerns capability. Through consolidated benchmarking the platform renders visible patterns of dispersion and anomaly that were always latent in the data but never accessible under the manual process.

The contribution of the study is that, where prior work has measured perceptions of accuracy without a spreadsheet baseline, or proposed design principles without validating them, this dissertation builds an operational system and measures it against the process it displaced. Its limits are equally clear. The controlled comparison covers three of the ten companies, the accuracy test relies on synthetic data, and the platform improves the fidelity of the reporting pipeline without conferring truth upon the figures entered into it. For the consortium, the practical value is a reporting cycle reduced by approximately two thirds and analyst effort redirected from routine preparation toward the assurance and interpretation that sustainability reporting increasingly demands.

1. Introduction and Background

1.1 Digital Transformation and the Evolution of ESG Reporting

The rapid growth of digital technologies has fundamentally transformed how organisations manage information and make strategic decisions in recent years. Cloud computing, advanced analytics, process automation, and artificial intelligence now enable businesses to process information at an unprecedented scale and speed, moving well beyond the constraints of traditional manual systems. However, it would be misleading to suggest this transition has been uniform or inevitable. While large enterprises increasingly embed digital tools into their core operations, many organisations, particularly those in regulated industries and complex data-reporting sectors continue to rely heavily on disconnected spreadsheet systems and manual workflows (Westerman et al., 2011; Amoako-Gyampah and Salam, 2004; Seddon et al., 2010).

Environmental, Social, and Governance (ESG) reporting exemplifies this disconnect between technological capability and organisational practice. Over the past decade, ESG has evolved from a peripheral corporate responsibility activity into a core strategic imperative. Investors, regulators, and stakeholders now demand transparent disclosure on environmental impact, labour practices, governance structures, and long-term organisational resilience (Elkington, 1997; TCFD, 2017; ISSB, 2023; United Nations, 2015). Yet despite this elevated importance, the majority of organisations still gather ESG data through spreadsheets, email exchanges, and manual consolidation processes, an approach that becomes increasingly fragile as reporting complexity grows (World Economic Forum, 2022).

This is not simply a case of organisations lagging behind technological adoption. ESG reporting presents genuine operational challenges that distinguish it fundamentally from financial reporting. Financial data flows through integrated accounting systems with standardised structures, but ESG metrics originate from diverse sources such as operational facilities, supply chains, human resources systems, energy meters, waste management contractors, and external certifications. Each source may use different units of measurement, reporting timelines, and calculation methodologies (Global Reporting Initiative, 2021). Consolidating this fragmented information into coherent narratives and standardised disclosures remains labour-intensive and error-prone, even in organisations with sophisticated data infrastructure.

This challenge is particularly acute within sustainability consulting and industry benchmarking initiatives. Consulting firms and industry associations such as the World Business Council for Sustainable Development (WBCSD) must collect, validate, and benchmark ESG data from hundreds of member organisations simultaneously. The operational burden of managing this data flow, identifying inconsistencies, and producing meaningful comparative insights falls squarely on spreadsheet-based workflows and manual coordination,

consuming resources that could be directed towards deeper analysis and strategic insight generation (WBCSD, 2021).

1.2 Why ESG Data Management Remains Challenging

The complexity of ESG data management arises from three structural factors. The first is the absence of the standardisation that financial reporting enjoys. Where accounting standards such as IFRS and GAAP furnish unified frameworks, ESG reporting remains fragmented across competing standards, among them the Global Reporting Initiative, the Sustainability Accounting Standards Board, the Task Force on Climate related Financial Disclosures and the emerging International Sustainability Standards Board. Each framework carries different scope definitions, boundary requirements and measurement specifications (ISSB, 2023), and organisations frequently map the same underlying data into several frameworks at once, which compounds the difficulty considerably.

The second factor is that ESG data collection is geographically dispersed and temporally fragmented. A multinational manufacturer aggregating greenhouse gas emissions must draw readings from facilities across many countries, each potentially using different energy accounting systems, conversion factors and reporting timelines. Water consumption may be managed by facility operations, waste by environmental departments, and employment data by human resources, with no central repository or shared taxonomy (World Economic Forum, 2022). Sustainability professionals consequently devote a substantial share of their time to reconciling data rather than analysing it.

The third factor is the concurrent pressure exerted by regulation and by stakeholder scrutiny. The EU Corporate Sustainability Reporting Directive now mandates detailed ESG disclosures for thousands of European companies, and the climate disclosure rules of the Securities and Exchange Commission mark a comparable shift towards mandatory climate reporting in the United States (SEC, 2024; European Union, 2022). Investors, meanwhile, increasingly evaluate disclosures for completeness and credibility, obliging organisations to demonstrate robust data governance and rigorous internal validation (Friede et al., 2015). Manual processes struggle to satisfy these dual demands for scale and reliability.

1.3 The Operational Challenge: Productivity Lost in Data Management

The operational impact of spreadsheet-dependent ESG reporting is substantial yet frequently underestimated. Sustainability professionals devote a considerable proportion of their working time to administrative tasks,

consolidating facility level submissions, reconciling discrepancies, correcting data entry errors and reformatting information for external frameworks and public disclosure platforms. Necessary though such tasks are for compliance, they are not the value adding work for which sustainability specialists are engaged. Their expertise is better directed towards analysis, strategic planning, benchmarking, and the generation of insight, which are the activities that drive organisational improvement.

More troubling is the quality of the data that emerges from manual processes. Manual workflows raise the likelihood of misplaced decimal points, inconsistent unit conversions, duplicate records, missing historical data and incomplete validation (English, 1999). Such errors carry material consequences. An organisation that reports incorrect emissions data, whether inadvertently or through control failures, may face regulatory scrutiny, investor disapproval or reputational damage once the error becomes public. This risk generates a perverse incentive, whereby organisations invest more effort in defensive validation than in developing their reporting.

Further difficulty concerns data governance and confidentiality. ESG datasets frequently contain commercially sensitive information about production volumes, facility locations, efficiency metrics, and competitive positions. When spreadsheets circulate across departments, through email and external consultants, they create audit and control problems. Organisations increasingly require role-based access controls, encryption, audit trails and approval workflows, none of which spreadsheets natively support (DAMA International, 2017). These governance gaps are not theoretical. Within industry benchmarking initiatives such as the WBCSD Tire Industry Project, data collection from member companies triggers a cascade of manual coordination, including validation of submissions, follow up communication to resolve inconsistencies, recalculation of missing time series and reconciliation across reporting cycles. This burden delays precisely the insight generation that participating companies value most (WBCSD, 2021).

1.4 Theoretical Foundations: Why Automation Might Help

Three bodies of literature suggest that automation could meaningfully address these operational challenges. The digital transformation literature demonstrates that organisations can improve efficiency and decision-making by redesigning processes around digital capabilities (Westerman et al., 2011; Fitzgerald et al., 2014). Business process automation research shows that repetitive, rule-based tasks are particularly amenable to automation, and that automated workflows can reduce processing time and error rates significantly (Davenport and Short, 1990; Hammer, 1990). Data quality management literature emphasises that systematic data validation, normalisation, and audit trails improve reporting reliability and reduce errors across reporting systems (Wang and Strong, 2015; Wang, 1998).

However, applying these theoretical insights to ESG reporting is not straightforward. ESG data quality problems arise not just from individual errors but from structural fragmentation as well. Automation alone cannot resolve fundamental measurement inconsistencies or governance ambiguities. Additionally, whilst digital transformation often prioritises efficiency gains, ESG reporting also demands stakeholder credibility and transparency. Automated systems must actively support these requirements through clear validation logic, transparent audit trails, and documented data transformations (World Economic Forum, 2022). This raises critical questions: Does automation improve efficiency specifically in ESG reporting, or does it merely shift effort from data entry to system configuration and troubleshooting? Does a platform reduce errors, or does it obscure them behind algorithmic processing? How can automated systems maintain the stakeholder confidence that manual, audited processes can provide? These questions motivate this dissertation research.

1.5 Research Gap: From Frameworks to Operations

Existing research on ESG reporting is extensive but concentrated in a particular direction: it focuses overwhelmingly on what organisations should disclose rather than on how those disclosures are operationally produced. Academic literature has thoroughly examined sustainable reporting frameworks, disclosure standards, and best practices in transparency (Global Reporting Initiative, 2021; ISSB, 2023) while regulatory and standards-setting bodies have invested heavily in defining ESG metrics, measurement boundaries, and assurance requirements. Technology and management literature has, in parallel, extensively documented the benefits of automation and digital transformation across industries.

What remains under-examined is the intersection of the two: how automation can improve the operational reality of ESG reporting, specifically the collecting, validating, consolidating, and analysing of ESG data within real organisations and consulting environments. Although some consulting firms and software vendors have developed proprietary platforms for ESG data management, peer-reviewed evaluation of these systems remains limited; the small number of academic studies that do exist (Pizzi et al., 2023; Münch et al., 2025) examine perceived disclosure quality or design principles rather than measured efficiency and accuracy gains. This gap matters because operational effectiveness directly determines whether organisations can meet regulatory requirements and stakeholder expectations in a resource-efficient manner. A further imbalance concerns organisational scale: existing digital-transformation and enterprise-systems research has concentrated on large organisations with mature IT infrastructures (Westerman et al., 2011; Seddon et al., 2010), leaving the ESG-reporting challenges of mid-market companies, industry consortia, and consulting firms, in which spreadsheets remain dominant, comparatively unexamined, despite being precisely where the operational burden is most acute.

1.6 Research Question, Aim, & Objectives

This research addresses this gap by investigating the following core question:

What impact does an automated ESG reporting platform have on the efficiency, accuracy, and productivity of sustainability reporting processes compared to traditional spreadsheet-based reporting systems?

The research aim is to evaluate the impact of an automated ESG reporting platform on sustainability reporting processes by measuring improvements in reporting efficiency, data accuracy, consistency, validation performance, and employee productivity compared to traditional spreadsheet-based reporting systems.

The specific research objectives are:

- To critically examine the challenges and limitations of traditional ESG reporting processes through a review of existing literature.
- To investigate the impact of manual ESG reporting processes on reporting efficiency, data quality, and employee productivity.
- To develop an automated ESG reporting platform capable of data collection, processing, validation, analysis, and benchmarking.
- To evaluate the impact of the platform on reporting efficiency, data quality, and employee productivity compared to traditional spreadsheet-based reporting systems.

1.7 Research Scope & Success Metrics

The success metrics applied in this evaluation were not devised ad hoc. Each derives from a construct established in the literature reviewed in Chapter 2, and each is operationalised in full in Section 3.4.2. In summary, the study assesses the platform against four criteria, namely reporting efficiency, data quality, analyst productivity and credibility.

Of these, credibility warrants emphasis at the outset, because an automated system that produces results faster but that stakeholders do not trust is operationally worthless. Platform design therefore emphasises transparency in validation logic, clear documentation of data sources and transformations, and comprehensive audit trails that allow stakeholders, whether internal or external, to understand and verify results.

2. Literature Review

2.1 The Operational Burden of Manual ESG Reporting

Before considering how automation might assist, the literature establishes why the problem matters. ESG reporting differs from financial reporting in ways that render spreadsheet-based management increasingly unsustainable. Hahn and Kühnen (2013) reviewed 178 articles spanning 1999 to 2011 and mapped the drivers of sustainability reporting adoption onto four theoretical lenses, namely legitimacy theory, stakeholder theory, signalling theory and institutional theory, identifying a persistent gap between frameworks and operational implementation. What has changed since 2013 is the scale of the problem. Regulatory frameworks, and the EU Corporate Sustainability Reporting Directive in particular, have transformed ESG reporting from a voluntary activity into a compulsory one, creating organisational pressure to systematise processes that were once ad hoc (European Union, 2022).

Investors, simultaneously, scrutinise the reliability of ESG data rather than merely its presence. The World Economic Forum's (2022) case studies of ten large reporting companies demonstrate that sustainability metrics originate in fragmented sources, among them facility operations, human resources systems, supply chain partners and external certifications, each employing different definitions, protocols and reporting timelines, which renders consolidation laborious and error prone.

The practical consequence is that sustainability professionals spend a disproportionate share of their time reconciling data rather than analysing it. This mirrors a longstanding productivity paradox in organisational information systems, whereby technology is adopted to improve efficiency but, if the underlying processes are not redesigned, merely accelerates an outdated workflow. The tension is captured in Hammer's (1990) influential argument that organisations should obliterate poor processes rather than simply automate them. It follows that establishing whether a platform improves efficiency requires an evaluation that first quantifies the administrative burden and then measures whether automation reduces it, which is the approach this dissertation adopts. Table A.5 in the appendix summarises every source reviewed, its methodology and the gap it leaves. The tables that follow each subsection below cover the studies discussed there.

Study	Method	Key findings and limitations
Hahn and Kühnen (2013)	Systematic literature review of 178 articles, 1999 to 2011	Maps sustainability reporting adoption onto legitimacy, stakeholder, signalling and institutional theory, and identifies a persistent gap between frameworks and operational implementation. Concerns why firms report rather than how reports are

Study	Method	Key findings and limitations
		produced, and pre-dates the shift to mandatory reporting.
European Union (2022)	Regulatory instrument, CSRD	Mandates detailed ESG disclosure for thousands of European firms, transforming reporting from voluntary to compulsory and creating pressure to systematise ad hoc processes. Establishes the obligation without addressing how firms are to meet it.
World Economic Forum (2022)	White paper, interview based case studies of ten reporting companies	ESG metrics originate in fragmented sources using differing definitions, protocols and timelines, with data historically collected by circulating spreadsheets. Diagnoses fragmentation without measuring its cost or testing a remedy.
Hammer (1990)	Conceptual argument, Ford case	Automation applied to an unreformed process merely accelerates an outdated workflow, hence the injunction to obliterate rather than automate. Illustrative rather than measured, and offers no framework for quantifying the burden it describes.

Table 2.1 - The Operational Burden of Manual ESG Reporting

2.2 Digital Transformation & the Adoption of ESG Reporting Platforms

The most directly relevant empirical evidence comes from recent studies of digital ESG-reporting platforms. Pizzi, Mastroleo, Venturelli and Caputo (2023) published one of the few peer-reviewed evaluations of how digital platforms affect sustainability-reporting quality. Using a fuzzy expert system grounded in contingency theory, they analysed Italian FTSE MIB companies and found that adoption of Sustainable Enterprise Resource Planning (S-ERP) systems led analysts to more accurate evaluations of environmental disclosure. The study demonstrates an improved perception of accuracy when data flows through a platform, but it does not measure operational efficiency, nor does it benchmark platform outputs against spreadsheets, which is the gap this dissertation occupies.

These platform specific findings sit within a broader body of work on digital transformation that establishes both the promise of such systems and the conditions under which that promise fails. Across large enterprises and ERP settings, this literature consistently finds that the gains from digital adoption follow from redesigning processes rather than layering technology onto them, and that the barriers to realising those gains are

organisational rather than technical (Westerman et al., 2011; Fitzgerald et al., 2014; Seddon et al., 2010; Amoako-Gyampah and Salam, 2004). Two implications follow. A platform must be evaluated on whether it reorganises work rather than merely digitises it, and, since this evidence derives almost entirely from large organisations with mature information technology functions, the reporting challenges of mid-market firms, consultancies and industry consortia remain comparatively unexamined, notwithstanding that spreadsheets persist most stubbornly there.

Münch, Mayer and Quick (2025) took a complementary approach using Design Science Research across two implementation cycles in a single organisation. They derived ten design principles for a harmonised ESG reporting platform that simultaneously satisfies external compliance requirements and internal sustainability management-accounting needs. Their design emphasised cross-functional collaboration and unified data structures bridging sustainability and financial information. This work is methodologically rigorous but limited by its single-case scope as it offers prescriptive design guidance rather than empirical validation of performance outcomes. The dissertation can extend this by testing whether their design principles, when implemented, produce the efficiency and accuracy gains they theoretically promise.

At the practitioner level, organisations including Workiva, SAP, and Plan A have developed commercial ESG-reporting platforms, yet their performance claims remain largely unvalidated in academic literature. This absence is itself significant mainly because millions of pounds are invested in platform adoption based on vendor claims, yet rigorous independent measurement of impact remains sparse.

Study	Method	Key Finding and Limitation
Pizzi, Mastroleo, Venturelli and Caputo (2023)	Fuzzy expert system grounded in contingency theory, Italian FTSE MIB companies	Adoption of Sustainable Enterprise Resource Planning systems leads analysts to more accurate evaluations of environmental disclosure. Measures perceived accuracy rather than operational efficiency, and does not benchmark platform output against spreadsheet workflows.
Münch, Mayer and Quick (2025)	Design Science Research across two implementation cycles in a single organisation	Derives ten design principles for a harmonised ESG reporting platform serving external compliance and internal management accounting needs. Prescriptive rather than evaluative, with no empirical validation that the principles deliver the promised performance.

Study	Method	Key Finding and Limitation
Westerman, Bonnet and McAfee (2011)	Multi case research across large enterprises	Gains follow from redesigning processes around digital capability rather than layering technology onto them. Concentrates on large firms with mature information technology functions.
Fitzgerald et al. (2014)	Survey and interview study of managers and executives	Organisational rather than technical barriers determine whether digital adoption succeeds. Broad in scope and not specific to reporting workflows.
Seddon, Calvert and Yang (2010)	Multi project benefits realisation model, case based	Most factors determining whether enterprise systems deliver benefits are organisational. Reinforces the large organisation bias in the evidence base.
Amoako-Gyampah and Salam (2004)	Survey based structural equation modelling in an ERP environment	Perceived usefulness and shared belief drive system adoption. Concerns user acceptance rather than measurable performance outcomes.
Commercial platforms: Workiva, SAP, Plan A	Vendor documentation and practitioner deployment, no peer-reviewed evaluation	Deployed across hundreds of organisations on the strength of vendor claims that remain largely unvalidated in academic literature.

Table 2.2 - Digital Transformation and the Adoption of ESG Reporting Platforms

2.3 Business Process Automation & Efficiency Gains

A consistent theme across the business process literature is that efficiency gains depend less on automation than on redesigning the work being automated. Hammer's (1990) reengineering argument, drawn from the Ford Motor Company case, established that radical redesign can yield dramatic labour savings, but stressed that the gain comes from obliterating a process rather than mechanising it. Davenport and Short (1990) formalised the same insight by distinguishing transactional, informational, analytical and transformational uses of information technology, finding across nineteen companies that the largest gains accrue where technology is used transformationally. Read together, and as summarised in Table 2.3, these works imply that an ESG platform which merely replicates spreadsheet routines will yield only modest gains, whereas substantial improvement requires rethinking how data flows, who validates it and when analysis occurs.

More recent work on Robotic Process Automation (RPA) supplies concrete evidence of these gains while sharpening the conditions under which they materialise. Drawing on Task-Technology Fit and Technology-to-Performance Chain theory, Kokina and Blanchette (2019) documented through interviews with adopters in accounting and finance that RPA reduced error rates, lowered costs, improved process documentation, and enforced standardisation but only where adoption was accompanied by genuine process redesign and by new oversight structures for managing exceptions, audit trails, and control logic. Schlegel et al. (2024) reinforce this caveat at the project level, finding that technical implementation accounts for only around 30% of the variance in RPA and business-process-automation success, with the remainder attributable to organisational and governance factors. The collective implication is that a platform cannot be evaluated in isolation: efficiency, governance alignment, and stakeholder credibility must be assessed together, since a faster system that weakens oversight or erodes trust may represent no real improvement over the spreadsheet processes it replaces.

Building on this synthesis, the dissertation adopts Davenport and Short's (1990) transactional-versus-transformational distinction as its primary evaluative lens, because it offers a principled basis for separating a platform that merely accelerates spreadsheet routines from one that enables new analytical capabilities, which is precisely the distinction on which the literature argues the value of automation depends. For the same reason and following Schlegel et al.'s (2024) finding that the majority of implementation success is organisational rather than technical, the evaluation deliberately measures not only platform efficiency but also the accompanying changes in roles, workflows, and governance; to omit these would risk attributing to the software outcomes that in fact depend on how it is embedded in practice.

Study	Method	Key Finding and Limitation
Hammer (1990)	Conceptual argument built on the Ford Motor Company accounts payable case	Radical process redesign reduced the department from over 500 staff to 125 clerks, a 75 per cent reduction, achieved by obliterating the existing process rather than mechanising it. Illustrative rather than measured, supplies no framework for quantifying the gains it describes.
Davenport and Short (1990)	Comparative case analysis across nineteen companies	Distinguishes transactional, informational, analytical and transformational roles for information technology, finding that the largest gains accrue where technology is used transformatively. Predates modern automation technologies. Adopted here as the primary evaluative lens.
Kokina and Blanchette (2019)	Interview study of RPA adopters in	Robotic process automation reduced error rates, lowered costs, improved documentation and enforced

Study	Method	Key Finding and Limitation
	accounting and finance, framed by Task Technology Fit and Technology to Performance Chain theory	standardisation, but only where accompanied by genuine process redesign and new oversight structures for exceptions and audit trails. Confined to accounting and finance, and qualitative rather than quantified.
Schlegel, Rosenberg, Fundanovic and Kraus (2024)	Synthesis of key success factors across RPA and business process automation projects	Technical implementation accounts for only around 30% of the variance in project success, the remainder being organisational and governance related. Concerns project success in general rather than reporting quality.

Table 2.3 - Business Process Automation and Efficiency Gains

2.4 Data Quality Management & Accuracy

If efficiency constitutes one pillar of the research question, data accuracy constitutes the other. The foundational literature is the data quality management framework of Wang and Strong (2015) and Wang's (1998) Total Data Quality Management cycle. Through a two-phase sorting study conducted with data consumers, Wang and Strong identified a hierarchical construct comprising four categories of quality, namely intrinsic, contextual, representational and accessibility quality, each resolving into a set of specific dimensions that are set out in full in Appendix A. This framework has become the canonical reference for operationalising data quality, and it applies directly to ESG data, where the pertinent questions are whether a platform improves the accuracy of reported metrics, the completeness of time series, the consistency of units and definitions across the reporting base, and the timeliness of the resulting insight.

Wang's TDQM cycle, which proceeds through the stages of define, measure, analyse and improve, is inherently process oriented and aligns with the traditions of continuous improvement and root cause analysis. Applied to ESG reporting, it implies that an automated platform ought to embed explicit validation rules, error detection and feedback loops. The limitation of the framework is its generality. It does not address measurement challenges specific to ESG, such as ambiguity over scope boundaries in greenhouse gas accounting, unit conversion across energy types, or aggregation across multiple facilities.

A more direct quantification of the ESG data quality problem comes from Berg, Kölbel and Rigobon (2022), whose decomposition of six major ESG rating agencies found that measurement methodology accounts for 56% of rating divergence, scope definition for 38% and weighting for only 6%, with pairwise correlations

between raters ranging from 0.38 to 0.71. The economic consequences are considerable, since firms are rated inconsistently by different providers, which breeds confusion and creates incentives to game the system. The finding points to the underlying data problem, that the same firm's environmental impact may be measured in several legitimate ways and yield incompatible conclusions. An automated platform, designed transparently with clear measurement definitions and validation rules, could mitigate this fragmentation within an organisation or a benchmarking consortium, although it cannot resolve definitional disagreement across an industry.

The dissertation accordingly operationalises data accuracy through Wang and Strong's multi-dimensional framework rather than a simplistic error count, because a single error rate would obscure the distinct ways in which ESG data can be deficient. Figures may be individually correct yet incomplete across the required time series, internally consistent yet incomparable across companies, or accurate yet delivered too late to inform a decision. Measuring accuracy, completeness, consistency, and timeliness separately therefore captures the full spectrum of data quality and permits the evaluation to identify precisely where automation improves reliability.

Study	Method	Key finding and limitation
Wang and Strong (2015)	Two phase sorting study conducted with data consumers	Establishes canonical data quality construct across intrinsic, contextual, representational and accessibility categories. General purpose, and silent on ESG specific problems such as scope boundaries and unit conversion.
Wang (1998)	Conceptual framework, product perspective on data	Proposes the define, measure, analyse and improve cycle. Process oriented and generic, prescribing validation without specifying it for any domain.
Berg, Kölbel and Rigobon (2022)	Quantitative decomposition of six rating agencies	Measurement methodology drives 56 per cent of ESG rating divergence, scope 38% and weighting 6%. Analyses divergence between external raters, not within a reporting organisation.

Table 2.4 - Data Quality Management and Accuracy

2.5 ESG Disclosure Quality & Standards Adoption

Beyond data quality considerations, the literature on disclosure quality and standards adoption supplies context for why comparability matters. Christensen, Hail and Leuz (2021) conducted an economic analysis and

literature review of mandatory corporate social responsibility and sustainability disclosure across jurisdictions. They found that mandates do improve the quantity of disclosure and widen the population of reporting firms, but that their effects on comparability and reliability are mixed and dependent on context. Mandates paired with standardised frameworks and assurance requirements improve reliability, whereas those imposed in the absence of harmonised standards merely proliferate divergent approaches.

Tawiah and Borgi (2022) examined the adoption of XBRL in financial reporting and found that structured data formats improve information efficiency, comparability and display quality, with the strongest effects observed in developing countries. The precedent is instructive for ESG. XBRL demonstrated that standardising the format of reported data, moving from narrative or proprietary spreadsheets to machine readable structures, yields measurable gains in comparability. A platform that enforces standardised metric definitions and formats could deliver analogous benefits, and the present dissertation tests whether the standardised definitions its platform enforces improve comparability across the organisations within the benchmarking consortium.

Study	Method	Key finding and limitation
Christensen, Hail and Leuz (2021)	Economic analysis and literature review across jurisdictions	Mandates improve disclosure quantity, but their effects on comparability and reliability are mixed and context dependent. Concerning market level regulation, not the systems producing disclosures.
Tawiah and Borgi (2022)	Global panel study of XBRL adoption	Structured data formats improve information efficiency, comparability and display quality. Concerns financial rather than ESG reporting, so offers precedent rather than direct evidence.

Table 2.5 - ESG Disclosure Quality and Standards Adoption

2.6 Automation in Accounting & Reporting Contexts

Research on robotic process automation and artificial intelligence in accounting offers the methodological precedents most analogous to the evaluation of an ESG platform. This literature indicates that automated text processing can assist in assessing disclosure quality and in flagging outliers and inconsistencies, yet it cautions consistently that trustworthiness remains the binding constraint, since stakeholders must be able to understand why an algorithm flagged or altered a disclosure.

Kokina and Blanchette (2019) reached the same conclusion in an accounting setting, finding that robotic process automation improved standardisation and documentation only where new oversight structures were introduced to manage exceptions and audit trails.

These findings align with Schlegel et al. (2024), who emphasised that the success of automation depends upon governance alignment. The methodological implication is direct. A platform cannot be evaluated purely on speed or error reduction. The evaluation must also examine whether internal analysts, external auditors and investors trust the platform's data and the logic that produced it, and the present study therefore measures stakeholder credibility alongside technical performance.

Study	Method	Relevance to this subsection
Kokina and Blanchette (2019) (see Table 2.3)	Interview study of RPA adopters	Automated processing improves standardisation but demands new governance for exceptions and audit trails.
Schlegel et al. (2024) (see Table 2.3)	Synthesis of automation success factors	Automation success depends on governance alignment, so speed and error reduction alone cannot evaluate a platform.

Table 2.6 - Automation in Accounting and Reporting Contexts

2.7 Multi Organization Data Sharing & Consortia

The context of this dissertation, the WBCSD Tire Industry Project benchmarking consortium, introduces a dimension largely absent from the literature reviewed above, namely the coordination of data across multiple organisations. No peer-reviewed management study of the Tire Industry Project's ESG data management process appears to exist, and the project's academic outputs concern tyre road wear particles rather than consortium governance.

Related work on inter organisational collaboration nonetheless offers insight. Nezami, de Bruijne, Hertogh and Bakker (2022) conducted a systematic literature review and interview study on data sharing in inter organisational infrastructure projects, finding that although the theoretical benefits of collaboration are well established, their practical realisation is severely hampered by weak data sharing governance. Information asymmetries, misaligned incentives and poor standardisation undermine collaboration, and ESG consortia plausibly face the same barriers. A platform designed to standardise data structures, enforce validation rules and create transparent audit trails addresses these governance gaps directly.

Veleva, Hart, Greiner and Crumbley (2003) benchmarked sustainability indicators across six pharmaceutical firms and found that most reported only eco efficiency metrics, rarely extending to life cycle or carrying capacity indicators, while Veleva and Ellenbecker (2001) proposed a multi level indicator framework intended to enable comparison across firms. Together these studies illustrate that even where the willingness to

benchmark exists, harmonisation of measurement is difficult and is rarely achieved voluntarily. This dissertation addresses the resulting gap in consortia research by evaluating whether an automated platform can overcome the governance and standardisation barriers that Nezami et al. and Veleva et al. identified.

Study	Method	Key Finding and Limitation
Nezami, de Bruijne, Hertogh and Bakker (2022)	Systematic review with practitioner interviews	Collaboration benefits are hampered by weak data sharing governance, information asymmetry and poor standardisation. Set in infrastructure construction, not ESG.
Veleva, Hart, Greiner and Crumbley (2003)	Benchmarking of six pharmaceutical firms, five level indicator hierarchy	Most firms reported only eco efficiency indicators, rarely extending to life cycle measures. Small sample, single industry and dated.
Veleva and Ellenbecker (2001)	Conceptual framework development	Proposes a multi-level indicator framework enabling cross firm comparison. Prescriptive, with no evidence of consistent adoption.

Table 2.7 - Multi Organisation Data Sharing and Consortia

2.8 Critical Synthesis

Read as separate domains, the literature reviewed above offers little guidance. Read together, it reveals both tensions, where different research streams recommend conflicting approaches, and complementarities, where distinct frameworks reinforce one another and are best used in combination.

Three tensions are salient. The first sets speed against credibility. The automation literature emphasises cost reduction and time saving as the primary benefits, whereas the data quality and governance literature emphasises accuracy, reliability and stakeholder trust. A platform that is 10 times faster but less transparent about its validation logic may prove operationally inferior to spreadsheets, because investors, auditors and boards will discount data from a system they cannot verify. This dissertation addresses the tension by measuring efficiency gains and credibility together, recognising that a successful platform must deliver both.

The second tension sets standardisation against flexibility. Platforms enforce standardised metric definitions to improve comparability, thereby addressing the measurement divergence that Berg et al. (2022) quantified. Yet organisations sometimes have legitimate reasons to measure sustainability metrics differently, reflecting facility characteristics, regional regulation, or business model. Rigid standardisation may improve comparability across organisations at the cost of validity for individual firms, and this dissertation documents where enforced standardisation improves outcomes and where it generates friction.

The third tension sets automation against process redesign. Hammer's (1990) warning that organisations should obliterate rather than automate retains its force. An organisation that merely replicates spreadsheet logic in software may gain speed while forgoing the opportunity to rethink the process itself, for instance by shifting from annual retrospective reporting to continuous monitoring. The comparison undertaken here establishes whether platform adoption prompted genuine rethinking or merely accelerated existing routines.

Two complementarities are equally instructive. Task-Technology Fit theory, invoked by Kokina and Blanchette (2019), explains which platform features correlate with improved performance, while Wang and Strong's (2015) framework specifies how data quality should be measured. Employed together, they require the evaluation to establish both whether the platform's features fit the reporting task and whether those features improve data quality across its full spectrum. Davenport and Short (1990) and Schlegel et al. (2024) similarly hold that the success of automation depends upon organisational adaptation and governance alignment, so this dissertation treats changes in roles, approval workflows, audit mechanisms and skill requirements not as incidental side effects but as core evidence of whether genuine process improvement occurred.

3. Methodology

3.1 Introduction & Chapter Overview

Chapter 2 concluded by identifying a clear gap. The literature establishes what organisations should report and demonstrates, in broad terms, that automation can improve business processes; what it seldom does is measure what changes when a real ESG-reporting workflow is automated and set against the manual process it replaces. This chapter sets out how the present study addresses that gap. It is organised around the research question posed in Chapter 1, which asks what impact an automated ESG-reporting platform has on the efficiency, accuracy and productivity of sustainability reporting when compared with traditional spreadsheet-based reporting.

The strategy is easily stated. The platform is built as a working system and then evaluated directly against the manual process it was designed to replace. In terms of the objectives, the chapter delivers RO2, RO3 and RO4, drawing on the literature synthesised under RO1 as the source of every measure it applies.

The chapter proceeds in four movements, following the structure expected of a methods chapter. Section 3.2 establishes what the study proposes to do to meet its objectives, and why a design-and-comparison strategy is the appropriate one. Section 3.3 describes the data the platform uses and how that data is processed. Section 3.4 details the methodology in two parts: the artefact itself, and the way its impact is measured. Section 3.5 connects the approach back to the literature of Chapter 2 and forward to the outputs that answer the research question, and Section 3.6 summarises the chapter. Ethics and governance are treated at Section 3.4.3 and reliability, validity and limitations at Section 3.4.4, both within the detailed methodology. Technical details that support but need not interrupt the argument are provided in the Technical Appendix.

3.2 Research Approach: Proposed Study to achieve the Objective

The proposal is direct. The platform is treated as the object of the research, developed as a working system, and then evaluated by comparing it against the manual process it replaces. This section sets out the reasoning behind that choice, namely the strategy selected, its justification over the alternatives that were considered, and the way that strategy maps onto the four research objectives.

3.2.1 Design Science Research & before and after comparison

The strategy adopted is Design Science Research (DSR), articulated by Hevner et al. (2004) and formalised as a staged process by Peffers et al. (2014). DSR is designed for research that produces and evaluates a purposeful

artefact addressing a real problem, which is precisely the nature of this project. It was selected over the obvious alternatives for concrete reasons. A controlled experiment was unworkable, since the study concerns a single real system in a single real setting rather than a large sample under laboratory conditions. A survey or interview study would have captured opinions about ESG platforms but not hard evidence of whether one works, which is the crux of the question. A conventional case study would describe the platform without committing the research to building it and testing it against a defined baseline, which DSR requires. DSR also connects the work directly to the literature: Münch et al. (2025) employed DSR to derive design principles for an ESG platform but stopped at the design stage, whereas this dissertation implements principles of that kind in a system that genuinely runs and supplies the evaluation their study could not.

Peppers et al.'s (2014) six activities structure the project, running from problem identification and the definition of solution objectives, both established in Chapters 1 and 2, through the design and building of the artefact in Section 3.4.1 to its demonstration and evaluation in Section 3.4.2, and finally to communication through this dissertation and the platform's continued use within dss+.

Design Science Research is inherently iterative, proceeding through successive cycles in which an artefact is built, evaluated and refined rather than produced in a single pass (Peppers et al., 2014). This project developed across three principal iterations. The first built a working platform against a synthetic dataset, which allowed the conversion and calculation logic to be exercised safely before any real data was involved. The second refined the data schema and the conversion engine in light of that testing, notably the three-column unit structure that emerged as the means of reconciling heterogeneous corporate submissions. The third migrated the matured platform to real client data and deployed it on Microsoft Azure. The methodology, evaluation and results reported in this dissertation describe the final iteration, since it is the state in which the artefact was assessed, while the earlier iterations are the developmental path by which that state was reached.

Because evaluation is the stage at which design research most often falls short, it was structured deliberately using the Framework for Evaluation in Design Science (FEDS) of Venable, Pries-Heje and Baskerville (2016). FEDS distinguishes controlled, artificial evaluation from situated, naturalistic evaluation, and the present study employs both. The platform's accuracy is tested in a controlled manner against a synthetic dataset whose correct answers are known in advance, while its efficiency and productivity are judged naturalistically, by comparing the automated workflow against the manual spreadsheet-based process it was built to replace. Combining the two guards against the weakness of each, since a controlled test alone would be too artificial and a real world comparison alone too noisy. The study accordingly draws on both quantitative and qualitative evidence, because whether a platform improves reporting is partly a matter of quantities that can be counted and partly a matter of whether those who rely on the numbers trust them.

3.2.2 Mapping the design to the research objectives

The four objectives map cleanly onto this design, which is part of its justification. RO1, which examines the limitations of traditional ESG reporting through the literature, was the work of Chapter 2 and supplies the methodology with every measure it later applies. RO2, which investigates the impact of the manual process on efficiency, data quality and productivity, is met by characterising the baseline in detail: the WBCSD Tire Industry Project (TIP) KPI Collection Tool and the manual consolidation that formerly occupied two to three analysts. RO3, the development of a platform capable of collecting, processing, validating, analysing and benchmarking ESG data, is the artefact described in Section 3.4.1. RO4, the evaluation of the platform against the manual process, is the comparison presented in Section 3.4.2. In short, the methodology executes RO2, RO3 and RO4 in sequence, with RO1 furnishing the instruments of measurement.

3.3 Data and Data Processing

This section addresses two questions on which the evaluation depends: what data the platform uses, and how that data is transformed into the clean, consolidated figures it serves. The mechanics of the schema and pipeline are summarised here and set out in full in the Technical Appendix.

3.3.1 Data provenance: from synthetic build data to real client data

The platform was developed and tested on a synthetic dataset constructed to match the real WBCSD TIP schema field for field. It covers ten tyre manufacturers across the seventeen years from 2009 to 2025 with 170 company-year records in total, each described by approximately 224 fields, and it spans the full breadth of the TIP framework, from ISO 14001 certification, production, water, electricity and fuels to Scope 1 and Scope 2 emissions, waste, occupational health and safety, workforce diversity and science-based targets.

The use of synthetic data during development was a deliberate methodological choice rather than a convenience. It furnished a known correct answer against which the build could be tested, it removed any risk of exposing commercially sensitive member data while the system was still being written, and it permitted errors and edge cases to be introduced deliberately so that the platform's validation logic could be stress-tested. Recognisable manufacturer names such as Bridgestone, Goodyear and Michelin are retained as realistic labels, but the figures attached to them are entirely synthetic.

The screenshot shows the 'TIP ESG Platform' interface for a client named 'Bridgestone'. The left sidebar contains navigation links: Home, My Dashboard, My Records, Benchmarks, Reports, Submit Data (highlighted in green), and Settings. The main content area is titled 'Health & Safety' and includes three sections:

- 7. Health & Safety:** Site-level audit coverage. Total sites (H&S): 40. Sites with external H&S audit: 24. Sites with internal H&S audit: 23. External audit coverage (auto): 60.0%. Internal audit coverage (auto): 57.5%.
- 8. Diversity & Inclusion:** Total employees: 218. Total female employees: 91. Total Board of Directors: 9. Female Board of Directors: 2. % Female employees (auto): 41.7%. % Female BOD (auto): 22.2%.
- 9. Science-Based Targets (SBTs):** Number of companies -- enter 0 or 1 per field as applicable. Total with SBT: 40. Validated: 8. Committed: 21. Non-committed: 11.

A red 'Submit & Save Data' button is located at the bottom of the form.

Figure 3.1 - The People & Governance section of the data entry form, capturing health and safety, diversity and science-based targets data.

Once the platform was finalised, in the state described in this dissertation, it migrated to real client data. The client provided actual ESG data covering the same fields and the same span of years, so the transition required no change to the schema or the pipeline. Before that data went live it passed through three gates: the data-security controls protecting it in transit and at rest, authentication restricting each company to its own records, and a round of quality assurance and quality control (QA/QC) checking the incoming figures for completeness and consistency. The synthetic dataset therefore served to prove the platform correct, while the real client data is what the finalised platform now runs on.

3.3.2 Data structure & schema

The same data is held in two complementary layouts, and the distinction matters for both the platform and the evaluation. The first is a long-form table, holding one row for each combination of company, year and KPI. Mirroring the WBCSD tool, it amounts to just over 38,000 rows, and, because it stores one fact per row, scales gracefully as members and years accumulate. It is therefore the production database hosted in Azure SQL. The second is a wide table, holding one row per company year and one column per KPI across 421 columns which is less suited to scale but faster to read, and so serves the dashboards, charts and benchmarking. The two are written together from every submission and so never diverge.

The schema stores more than a single value per field because companies do not report in common units. The platform therefore defines a fixed common unit for each field and converts every submission to it, retaining both the corporate value, as the company entered it, and the converted common value, so that a company can view its records exactly as submitted while every calculation, chart and benchmark runs on a common basis across all ten companies. The full three column structure and a worked conversion example are given in Appendix B.

The synthetic build dataset carries one further column absent from the live data as a ground-truth value recording the exact common-unit figure each conversion should produce. Present only in the synthetic fixture, its sole purpose is to allow the platform's conversions to be tested against a known correct answer and the basis of the accuracy test in Section 3.4.2.3. The real client data has no such column, and its quality is instead assured through the QA/QC gate.

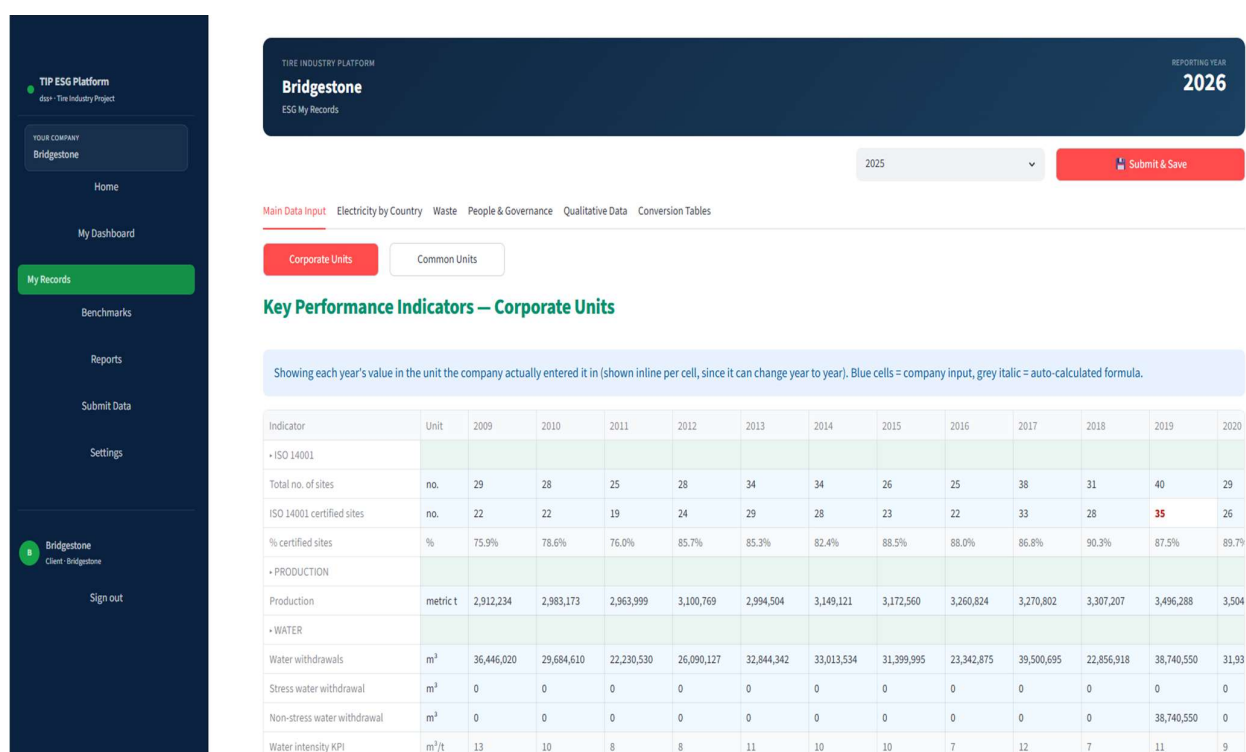


Figure 3.2 - Company records switchable between corporate and common units, company input values and unified unit values are distinguished.

3.3.3 The data processing pipeline & ground-truth validation

Processing transforms a raw submission into the consolidated dataset the platform serves, and the sequence is worth stating because the resulting audit trail is part of the contribution. On submission, every unit-bearing

field is converted to the fixed common unit using factors drawn directly from the WBCSD tool's own conversion sheets. The KPI engine then computes the derived intensities, totals and shares, and the long form and wide tables are written together so that they cannot drift apart. The sector averages and quartiles used by the benchmarking and analysis screens are computed live from the consolidated data rather than stored, so that a newly submitted year is reflected everywhere the moment it is saved.

The pipeline is designed to handle imperfect data, and its validation rules catch the kinds of errors a manual process tends to miss such as waste recovery exceeding total waste, certified sites exceeding total sites, or any figure moving more than a set threshold year-on-year. These rules resurface as flags in the analyst verification queue (Section 3.4.1), so that a problem detected during processing becomes a prompt for review rather than a silent error. Because the synthetic build dataset has a known correct state, the entire pipeline can be validated against it through a build routine that reconstructs the wide table from the long-form data, compares every converted value with its ground truth, and fails on any discrepancy beyond a rounding tolerance. In effect the data layer ships with its own accuracy test, which underwrites the accuracy claims made later, while the real client data, lacking an independent answer, is assured through the QA and QC gate. The handling of missing units and redundant columns is detailed in Appendix B.

3.4 Detailed Methodology

This section presents the methodology in two parts: the artefact, meaning the platform, described so that the subsequent evaluation can be trusted, and the evaluation itself, the way the platform is measured against the manual process it replaces. Ethics and limitations close the section. Component-level implementation details are provided in the Technical Appendix.

3.4.1 The artefact: the ESG Reporting Platform

The artefact at the centre of this research is a web-based ESG reporting and analysis platform, developed for the Tire Industry Project, a sustainability initiative convened under the World Business Council for Sustainable Development. It replaces a manual process, in which each member company completed its own copy of the WBCSD KPI Collection Tool by hand and analysts consolidated and checked those workbooks individually, with a single system that runs the entire cycle from data entry through validation, consolidation, benchmarking and reporting. The platform is written in Python with a Streamlit interface and is deployed on Microsoft Azure, using App Services for the application, Azure SQL for the long-form database and Blob Storage for documents and version snapshots. It was built and tested locally behind a storage-abstraction layer whose interface is identical to its Azure counterpart, so that migration to the cloud altered only that isolated layer rather than the complete application. The full implementation is detailed in Appendix B.

The platform presents a different face to each of its two users. A member company signs in to a client workspace built around its own data, with a dashboard, its records, its benchmarks and a submission form (Figure 3.3), while a dss+ analyst signs in to an internal workspace spanning the whole portfolio, with every company's data, a verification queue, cross-company analysis and sector reporting. This division matters for the evaluation, because the two roles correspond to the two halves of the manual process the platform replaces, namely the company that prepares a submission and the analyst who consolidates and scrutinises it. On submission, a company enters figures in whatever units its own systems use, and the platform converts each value to a fixed common unit in the background, displaying a running total as it does so (Figure 3.4).

Figure 3.3 - The platform entry point, where a user signs in as either a TIP client company or a dss+ analyst.

Figure 3.4 - The energy section of the submission form, with per field unit selection and a live common unit total.

A calculation engine then applies the TIP reporting rules, summing energy by source, applying IPCC emission factors to derive Scope 1 emissions and country-level IEA grid factors to derive Scope 2, and producing the intensity KPIs on which all comparison rests (Figure 3.5). Centralising unit conversion and calculation in this way removes an entire class of manual error and makes consistency a property of the software rather than of the analyst's memory.

Figure 3.5 - The fuels section, showing the granular coal and fuel oil sub types that mirror the WBCSD schema.

Before a submission is trusted, the platform validates it, flagging internal contradictions and year-on-year movements beyond a set threshold, and combining completeness with flag severity into a single readiness score that tells an analyst at a glance whether the submission is fit for the consolidated report (Figure 3.6).

Figure 3.6 - The dss+ verification queue, where year-on-year flags are accepted or queried before submission is approved.

Each save updates the wide and long-form tables together, regenerates the consolidated and per-company files, and writes an immutable timestamped snapshot, so that every figure carries a reviewable history. The consolidated data then drives the outputs, which include a company dashboard of performance and trends (Figure 3.7), a benchmarking view placing each company against the sector average and quartiles (Figure 3.8), and, for the analyst, a sector-wide analysis view aggregating all ten companies (Figure 3.9).

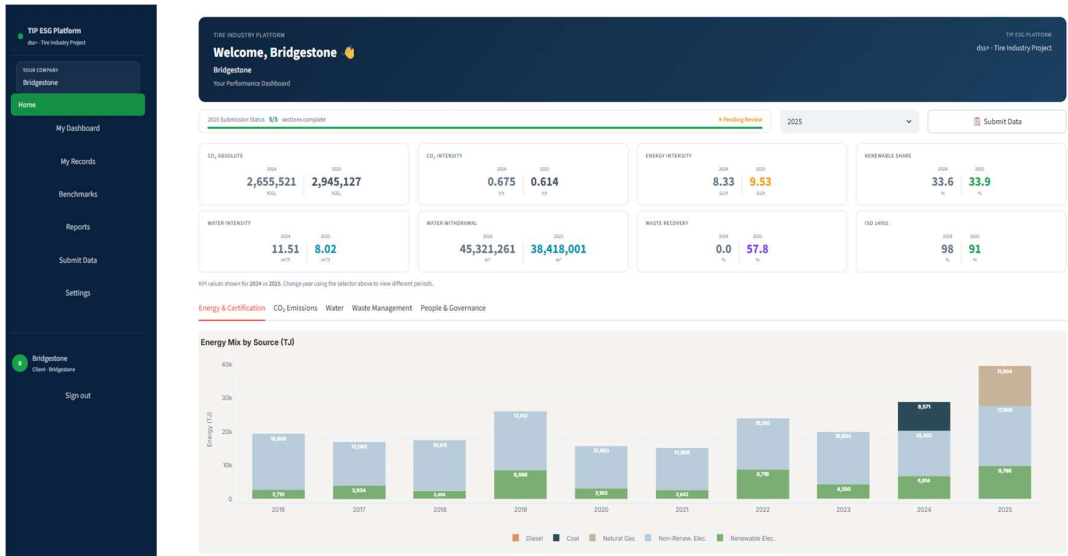


Figure 3.7 - A member company's performance dashboard.

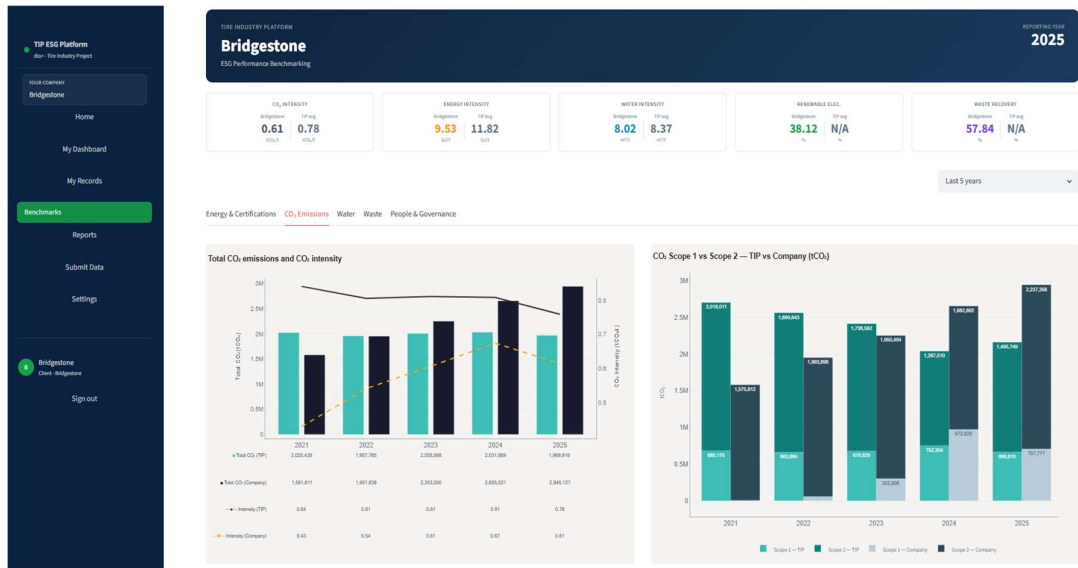


Figure 3.8 - Company versus sector benchmarking, including Scope 1 & Scope 2 emissions against the TIP aggregate.

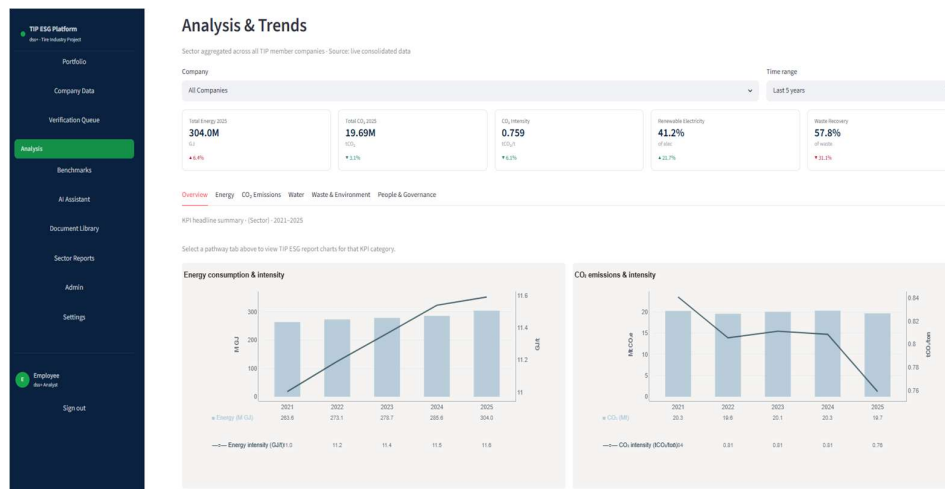


Figure 3.9 - The dss+ analysis and trends view, aggregating all ten companies into a sector picture.

None of these is a mere acceleration of the manual process; the sector aggregates and standing benchmarks are capabilities it did not possess. An AI assistant built on Azure OpenAI accompanies these screens but remains at an earlier stage of development and is treated here as a supporting feature and a direction for future work rather than as part of the evaluated pipeline.

3.4.2 Evaluation methodology

Having described what was built, this part sets out how its performance is measured by comparing the platform against the manual process it replaces across four dimensions, namely efficiency, data quality, productivity and credibility.

3.4.2.1 The comparison design: manual baseline versus platform

The evaluation poses a single question in four registers: does the platform perform the same reporting work as the manual process, but better? The comparison is accordingly a parallel run. The "before" is the manual workflow, in which member companies completed copies of the WBCSD KPI Collection Tool and the two to three dss+ analysts responsible for reporting consolidated those workbooks, reconciled units by hand, checked the figures and assembled the company and sector outputs. The "after" is the same work performed through the platform. The tasks compared are those the users actually performed on the company side, entering or updating a reporting year and then consulting the dashboards and benchmarks; on the analyst side, reviewing a company's data, benchmarking any company against the sector aggregate, and producing the consolidated report across all ten companies. Because the same task is performed both ways on the same data, the only variable that changes is the tool, and any difference in effort can reasonably be attributed to the platform. As this is an individual project, the manual baseline is measured directly for three companies through a timed

parallel run under both methods, while for the remaining seven it is reconstructed from analysts' informed estimates, stated openly as such.

3.4.2.2 Reporting efficiency

Efficiency is measured, following Hammer (1990) and Davenport and Short (1990), as the time and effort a full reporting cycle demands. Each of the five stages, collection, processing, validation, analysis and verification, is compared across the manual process and the platform on the effort it requires, namely analyst hours, the number of people involved, and the number of validation rounds before the data is accepted as clean. Because the earlier process was conducted by a small team over a known cycle, these quantities are expressed as approximate before and after figures and reported as such. The expectation drawn from Chapter 2 is specific, since a platform that merely moved the spreadsheet online would save little, whereas one that genuinely relocates conversion, calculation and consolidation into software should compress the cycle substantially.

3.4.2.3 Data quality and accuracy

Data quality is assessed through Wang and Strong's (2015) four dimensions, established in Chapter 2. Accuracy is measured directly, since the synthetic build dataset carries a ground truth for every field against which the platform's converted figures are checked, the round trip test of Section 3.3.3 performing this across the whole dataset. Completeness is the proportion of required fields populated, consistency is conformance to the TIP definitions and unit conventions that the engines enforce automatically, and timeliness is the interval between submission and the availability of validated data. Errors are characterised by number and severity through the platform's flags.

3.4.2.4 Analyst productivity

Productivity is assessed as a change in how analyst time is allocated. In the manual process the greater part of the analyst effort was absorbed by collecting, reconciling and checking data rather than interpreting it. The platform is intended to invert that balance, absorbing the routine load so that analyst time is released for analysis, benchmarking and advice. Productivity is therefore measured as the shift in where effort is directed, namely the reduction in preparatory work and the capacity that opens for higher value activity.

3.4.2.5 Credibility & governance

Chapter 1 identified credibility as a critical success factor, on the straightforward logic that a faster system whose numbers are not trusted is worthless. This dimension is assessed qualitatively, through the features that

render the platform's output defensible: the transparency of its validation logic, the traceable link between a company's corporate-unit entry and the common-unit figure used in analysis, the verification queue through which an analyst signs off each submission, and the comment-and-snapshot trail that records every edit. Where the manual process produced figures whose provenance resided in an analyst's memory and a tangle of workbooks, the platform renders that provenance an explicit and reviewable property of the system. It is weighed not as a measured quantity but as evidence that the gains in efficiency and accuracy are not achieved at the expense of trust.

3.4.3 Ethical considerations and data governance

Because the finalised platform runs real client data, governance is a substantive concern rather than a formality, and the platform was built to treat it as such. Access is role-based, so that each company reaches only its own records, and every figure is auditable through the snapshot and comment trail. The transition from synthetic to real data passed through data-security controls in transit and at rest, authentication, and a round of QA/QC before the data went live. Commercially sensitive ESG data is handled under dss+ governance and the applicable data-protection regime, including the General Data Protection Regulation where relevant. The AI assistant is constructed with the same caution. Only anonymised, derived KPI summaries ever leave the platform, company identities are stripped before any external call, and its output is treated as a draft requiring analyst review rather than a finished answer. During development, the use of synthetic data ensured that no real company information was exposed while the system was still being written.

3.4.4 Reliability, validity & limitations

The reliability of the evaluation rests on a fixed task set and a deterministic calculation engine, so that identical inputs always yield identical outputs and the accuracy test is reproducible by construction. Validity is supported on two fronts. Internally, data and tasks are held constant across the manual and automated conditions, so that differences may be attributed to the platform. At the level of construct, every measure is anchored in an established framework from Chapter 2 rather than an ad hoc definition, so that efficiency, data quality and productivity are measured as the literature defines them. The limitations of the approach are weighed against the findings themselves in Section 5.5.

3.5 Linking the Methodology to Literature Review & Research Question

The methodology is only as strong as its foundations. This section demonstrates that the approach did not arise arbitrarily, since every measure derives from the literature reviewed in Chapter 2, and every output of the

evaluation maps back to the objectives and the research question. The first part traces the measures to their sources, and the second traces the outputs forward to what they answer.

3.5.1 Linking the methodology to the literature review

Each element of the evaluation derives from the literature reviewed in Chapter 2, which is what lends it authority. The efficiency measure follows the process redesign tradition of Hammer (1990) and Davenport and Short (1990), framing efficiency as a question of whether the platform reorganises the work rather than merely relocating it. The data quality measure is taken directly from Wang and Strong (2015), whose four dimensions become the criteria by which the platform data is judged. The productivity measure operationalises the observation that administrative data work dominates analyst time. The attention paid to credibility and governance answers the cautions of Kokina and Blanchette (2019) and Schlegel et al. (2024) that automation delivers its benefits only where oversight adapts. The complete strategy extends Münch et al. (2025) from design principle to a running and evaluated system. The methodology is, in short, Chapter 2 rendered as a working instrument.

3.5.2 Linking analytical outputs to the objectives & the Research question

The outputs of the evaluation are constructed to answer the research question part by part. The question concerns three matters, namely efficiency, accuracy and productivity, automated against manual. The efficiency comparison and the productivity analysis together address the efficiency and productivity strands, furnishing RO2 with its baseline picture of the manual process and completing the corresponding parts of RO4. The accuracy test and the wider data quality measures address the accuracy strand and complete the data quality limb of RO4. The platform itself, by collecting, processing, validating, analysing and benchmarking in one system, is the output that satisfies RO3. Read back through the objectives, the fit is exact, as RO1 supplied the measures in Chapter 2, RO2 is met by characterising the manual baseline, RO3 by the platform, and RO4 by the comparison running through Section 3.4.2. Read back through the research question, the chapter is built to produce precisely the three comparisons it demands. The findings chapter reports those comparisons, and the present chapter has explained how they are produced and why they may be believed.

3.6 Chapter Summary

This chapter has set out how the study answers its research question. It took a pragmatic stance and chose Design Science Research as the strategy, explaining why a build and compare approach fits the question better than an experiment, a survey or a plain case study. It described the data the platform runs on, from the synthetic dataset used to build and test the system to the real client data the finished platform now uses, and it set out

how that data is converted, calculated and checked. It then detailed the platform component by component, and explained how its efficiency, accuracy, productivity and credibility are measured against the manual process it replaces. Finally, it tied every measure back to the literature in Chapter 2, and every output forward to the objectives and the research question. The next chapter applies this methodology and reports what the comparison shows.

4. Results

4.1 Introduction

This chapter presents the results of the evaluation set out in Chapter 3. It reports what the platform achieved, measured against the manual process it replaced, and interprets those findings in light of the literature reviewed in Chapter 2. The results are organised by research objective. The dataset and the manual baseline establish the conditions of the comparison, the data-quality, efficiency and productivity findings address RO4, and the analytical outputs speak to RO3.

All results reported here derive from the synthetic evaluation dataset, which is the system on which the platform was built, tested and formally assessed. This is a deliberate choice, since only the synthetic data carries the ground-truth values against which accuracy can be verified exactly. The finalised platform was subsequently migrated to real client data covering the same fields and the same period, which passed the security, authentication and quality assurance gates described in Section 3.3.1. That migration demonstrates transferability but is not itself the object of evaluation.

4.2 The Dataset & Its Characteristics

The evaluation dataset comprises ten tyre manufacturers reporting across seventeen years, from 2009 to 2025. This yields 170 company-year records described by 224 distinct KPI fields, amounting to 38,080 observations in long-form and 421 columns in the wide analytical layout (Table 4.1).

Dimension	Value
Companies	10
Reporting years	17 (2009 to 2025)
company-year records	170
Distinct KPI fields	224
Observations, long-form	38,080
Columns, wide analytical form	421
unit-bearing fields	23
ground-truth values available	3,910

Table 4.1: Scale and structure of the evaluation dataset

Coverage of the core reporting fields is complete. Production, water intake, total energy, total emissions and site counts are populated by every company in every year, at 100 per cent. Of the 421 wide columns, 194 are fully populated. Overall cell completeness stands at 46.6 per cent, but this figure is misleading if read naively, since 157 of the columns record electricity purchased by individual countries, and no manufacturer operates in every country in the taxonomy. Sparsity in those columns is a property of the schema rather than a deficiency in the data (Table 4.2).

Category	Fields	Populated	Note
Core reporting (production, water, energy, emissions, sites)	5	100%	Complete for every company and year
Energy by source	13	100%	Includes coal and fuel oil sub types
People & Governance (employees, board, health and safety, science-based targets)	13	100%	Complete
Electricity by country	157	Sparse by design	No manufacturer operates in every country in the IEA taxonomy
Waste	3	Approximately 60%	Waste recovery populated for one company and one year only
All wide columns	421	46.6% overall, 194 fully populated	Sparsity concentrated in country level electricity

Table 4.2: Completeness by field category

The most consequential characteristic of the dataset concerns units. Of the 23 fields that companies enter with a unit attached, every single one is reported in more than one unit across the 10 companies, at a mean of 5.9 distinct units per field (Figure 4.1). Production alone arrives in five units, namely metric tonnes, United States short tons, pounds, kilograms and tyres per year. Natural gas arrives in eight, spanning GJ on both higher and lower heating values, MWh on both, decatherms, kWh, TJ and British thermal units. Water arrives in four

(Table 4.3). A further 470 records carry a value with no unit recorded, roughly 12% of unit-bearing observations, and are handled by the documented fallback described in Section 3.3.3.

Field	Units used across the ten companies
Production	metric tonnes, US short tons, pounds, kilograms, tyres per year
Water intake	cubic metres, US gallons, litres, megalitres
Natural gas	GJ (HHV), GJ (LHV), MWh (HHV), MWh (LHV), kWh (HHV), TJ (LHV), BTU (LHV), decatherms (LHV)

Table 4.3 - Unit heterogeneity

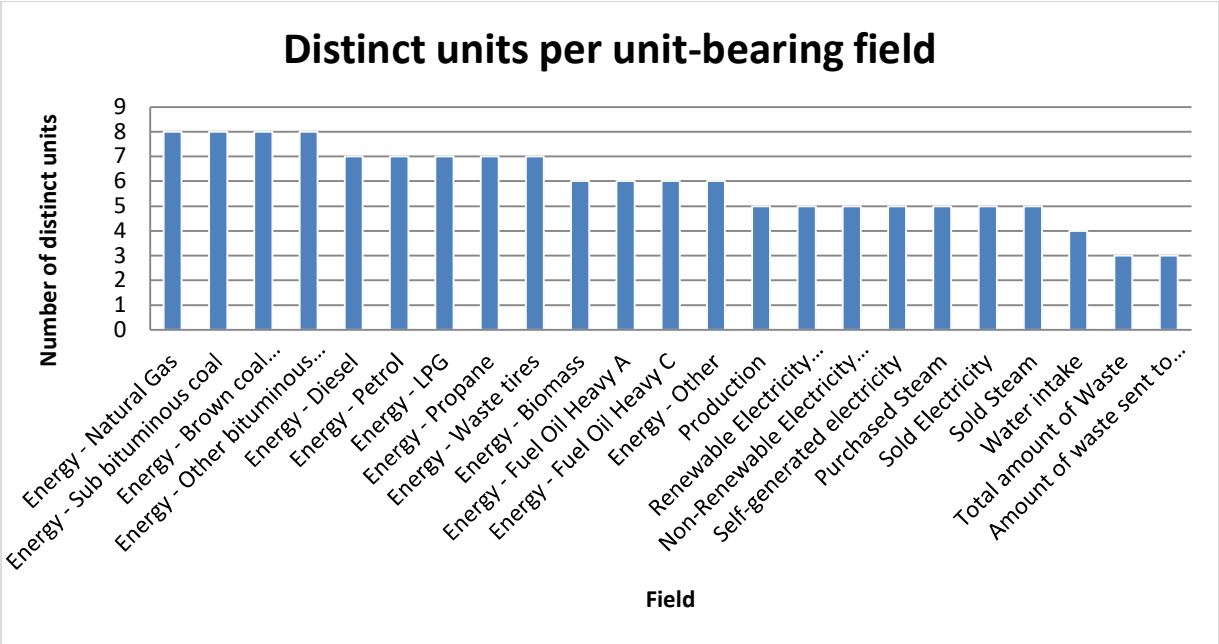


Figure 4.1 - Distinct units per field

This finding is not incidental. It establishes empirically, within a single benchmarking consortium, that the raw submissions of 10 organisations are not directly comparable in their reported form. Any consolidation that did not reconcile these units would produce a sector aggregate that is arithmetically meaningless.

4.3 The Manual Baseline

Before the platform, ESG reporting for the consortium was conducted by a small team of analysts working with the WBCSD KPI Collection Tool and a set of linked spreadsheets, with each cycle covering collection, processing, validation, analysis and verification for every member company. A full manual cycle for a single company was a substantial undertaking, and data was rarely accepted on first submission, typically requiring several rounds of validation and correspondence before a company's figures were considered clean.

For seven of the ten companies, the scale of that effort is known only from the analysts' own estimates, given in response to direct questions during the internship about how long each company's reporting consumed. These range from 24 to 32 hours per company (Figure 4.2). They are reported as recollected approximations rather than measured values, and the upper end of each estimate is used, so that the manual burden is not understated in the platform's favour. These figures are treated as standalone context and are not used in any direct comparison; the three companies for which both methods were measured are treated separately in Section 4.5.

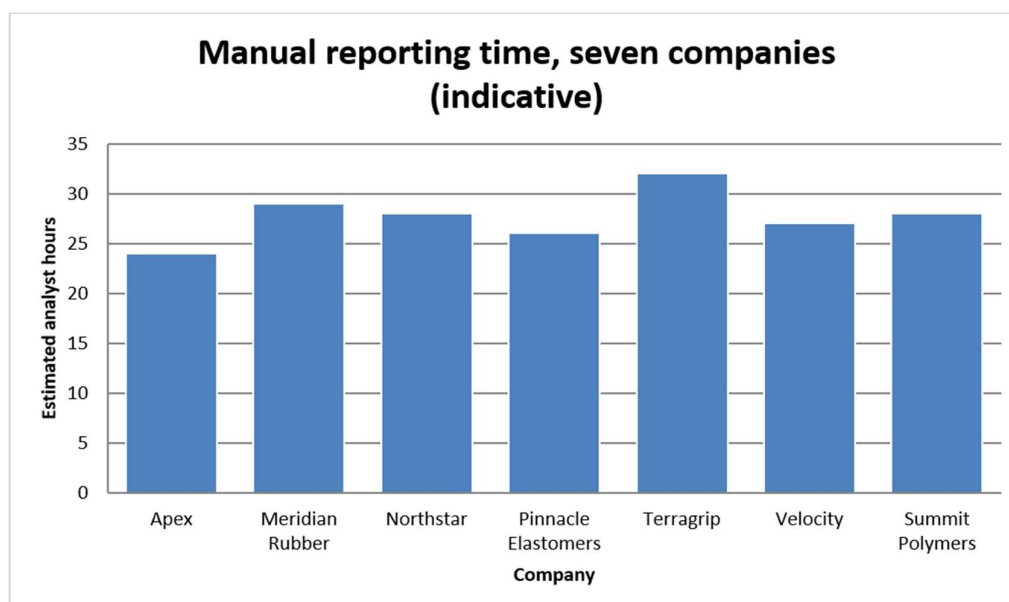


Figure 4.2 - Manual reporting time, seven companies (indicative)

4.4 Data Quality & Accuracy

4.4.1 Conversion accuracy against ground truth

The platform's unit conversion and calculation pipeline was tested against the known correct state of the synthetic dataset. The build routine reconstructs the wide analytical table from the long-form records, converts

every unit-bearing value to its common unit, and compares the result with the ground-truth figure that conversion should produce, failing on any deviation beyond a relative tolerance of one part in a million.

Of 3,910 conversions tested, 3,910 passed. There were no failures, and no unrecognised unit pairings. The pass rate is 100.00 per cent (Table 4.4).

Measure	Result
Unit-bearing fields tested	23
Company-year records	170
Conversions tested against ground truth	3,910
Passed within tolerance	3,910
Failed	0
Unrecognised unit pairings	0
Relative tolerance applied	1×10^{-6}
Pass rate	100.00%
Records with no unit recorded, handled by fallback	470 (12.0%)

Table 4.4 — Ground-truth conversion test

This result should be read carefully. It does not assert that the platform is incapable of error. It asserts that across every unit-bearing field, for every company, in every year, the conversion from a company's reported unit to the platform's common unit reproduces the correct value to within rounding. The equivalent assurance in the manual process did not exist, because no independent correct answer was ever computed against which a hand conversion could be checked.

4.4.2 A documented failure, its detection and its correction

During development, several records were discovered in which a company's corporate unit had been recorded incorrectly at the point of entry. A value entered in one unit was labelled as another. Because the conversion engine applies the factor associated with the recorded unit, the error propagates silently into the common unit

value, and from there into the intensity KPIs and the sector aggregates. The figures remained internally consistent and passed casual inspection. They were simply wrong.

These cases were detected precisely because the ground-truth column existed, and they were corrected at source. The episode illustrates a principle that the manual process could not embody. A conversion error arising from a mislabelled unit is invisible to inspection, since the output is a plausible number of the right order of magnitude. It becomes visible only when the system holds an independent expectation of what the answer should be. The platform's three column schema, storing the corporate value, the unit and the converted common value, is what made both the error and its correction possible.

4.4.3 Anomaly detection across the dataset

Applying the platform's validation rules across all 170 company-year records surfaced 97 anomalies (Table 4.5, Figure 4.3), comprising 51 instances of total energy moving more than 20% year-on-year and 46 instances of total emissions doing likewise. No structural errors were found, meaning that in no record did waste recovery exceed total waste, nor certified sites exceed total sites.

The distinction matters. Zero structural errors indicate that the dataset is internally coherent. The 97 warnings are not errors but flags, each representing a movement large enough to warrant an analyst's judgement as to whether it reflects a genuine operational change or a reporting mistake. In the manual process, each of these would have had to be noticed by eye across a spreadsheet spanning seventeen years. The platform surfaces them automatically in the verification queue (Figure 4.4), where an analyst accepts or queries each in turn.

Rule	Severity	Instances
Waste recovery exceeds total waste	Error	0
Certified sites exceed total sites	Error	0
Total energy moves more than 20% year-on-year	Warning	51
Total emissions move more than 20% year-on-year	Warning	46
Total anomalies surfaced		97

Table 4.5: Validation rules applied across all 170 records

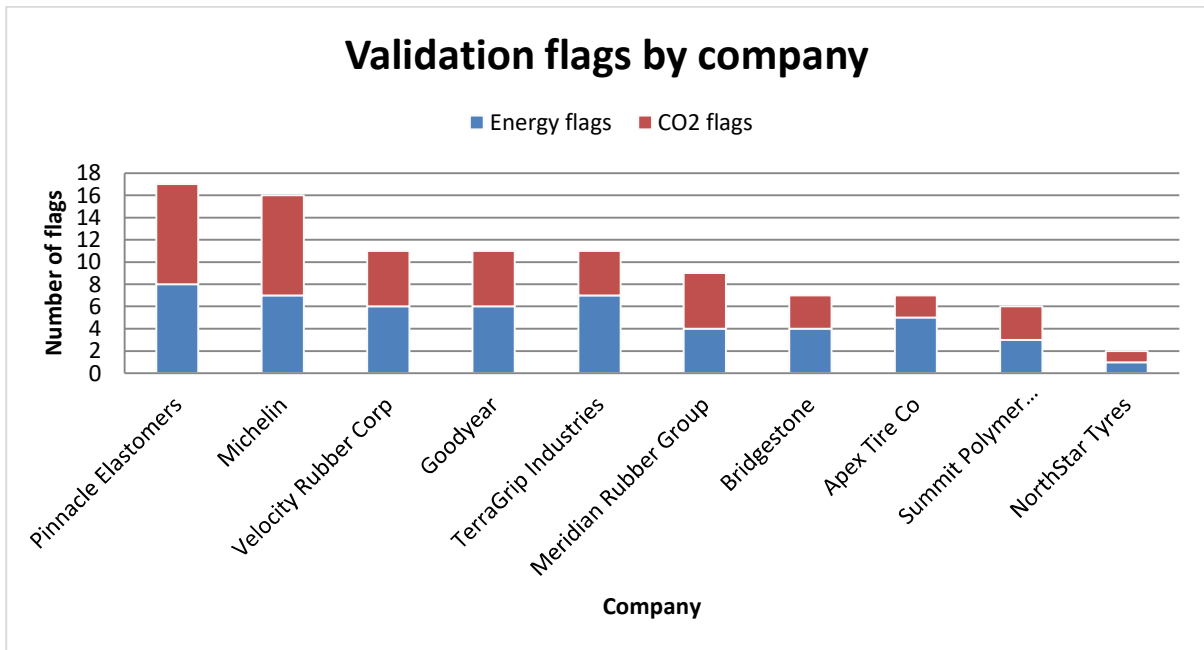


Figure 4.3 - Anomalies surfaced by company across all validation rules

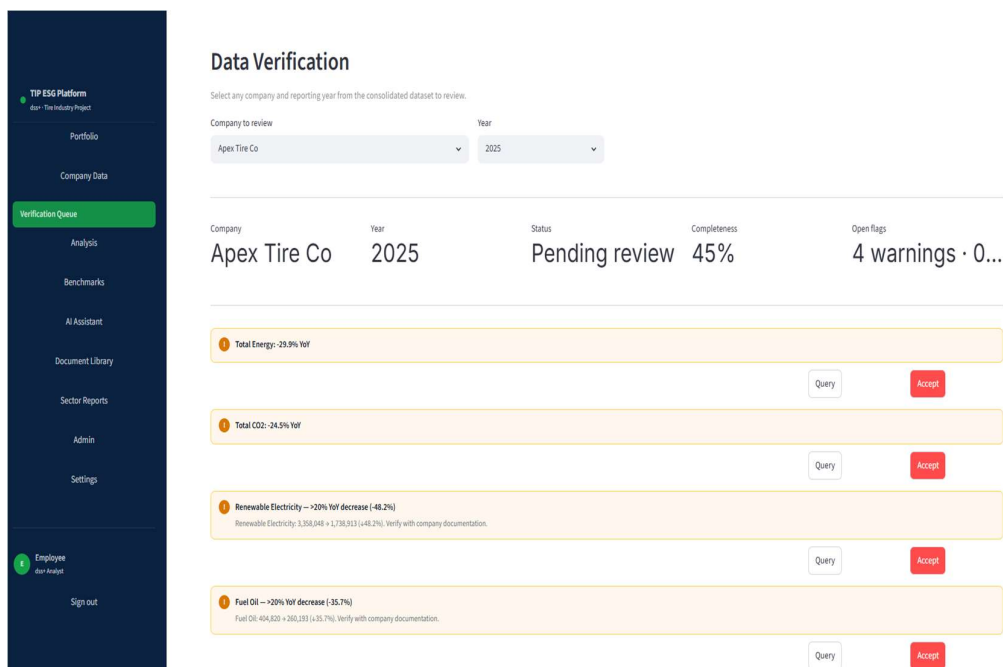


Figure 4.4 - The dss+ verification queue, where flagged anomalies are accepted or queried before approval.

4.5 Efficiency & Productivity

To measure the platform's effect rather than estimate it, three companies, Bridgestone, Goodyear and Michelin, completed a full reporting cycle under both methods after deployment, with each stage timed. The results are shown in Table 4.6 and Figures 4.5 to 4.7.

The pattern is consistent across all three. The manual cycle took between 18 and 21 hours per company, while the same work on the platform took 7 to 8, a reduction of between 61 and 65% (Figure 4.5). More telling than the total is where the hours went. On the platform, unit conversion, KPI calculation and consolidation are performed automatically on save: every unit-bearing field is converted to the common basis, the derived intensities and totals are recomputed, the consolidated and per-company files are regenerated, and the sector aggregates refresh. In consequence the two stages the platform absorbs into software, collection and analysis, nearly vanished, falling in aggregate from 6 hours and 19 hours to roughly 1 and 2 (Figure 4.7).

What the platform does not remove is the human judgement the process genuinely requires. Validation and verification remain the substantive stages, together accounting for the majority of the platform's residual time, because the system flags anomalies but an analyst must still adjudicate them and approve the result. This is the productivity shift the platform is designed to deliver as it absorbs the preparatory work of collecting, converting and consolidating data, and leaves the analyst's time concentrated on the checking and sign-off where expertise is actually exercised. The effect is not simply that reporting became faster, but that the balance of effort moved from preparation toward assurance.

Company	Manual	Platform	Hours saved
Bridgestone	21	8	13
Goodyear	20	7	13
Michelin	18	7	11

Table 4.6 - Total reporting effort per company in the three company parallel run (hours)

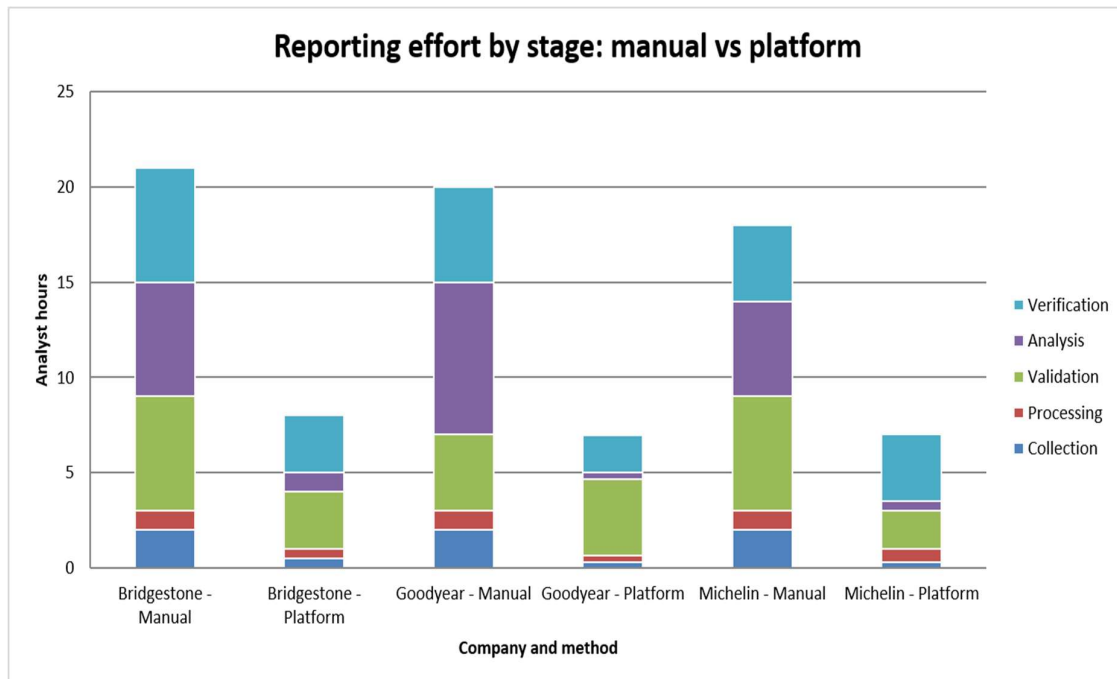


Figure 4.5 — Reporting effort by stage: manual versus platform

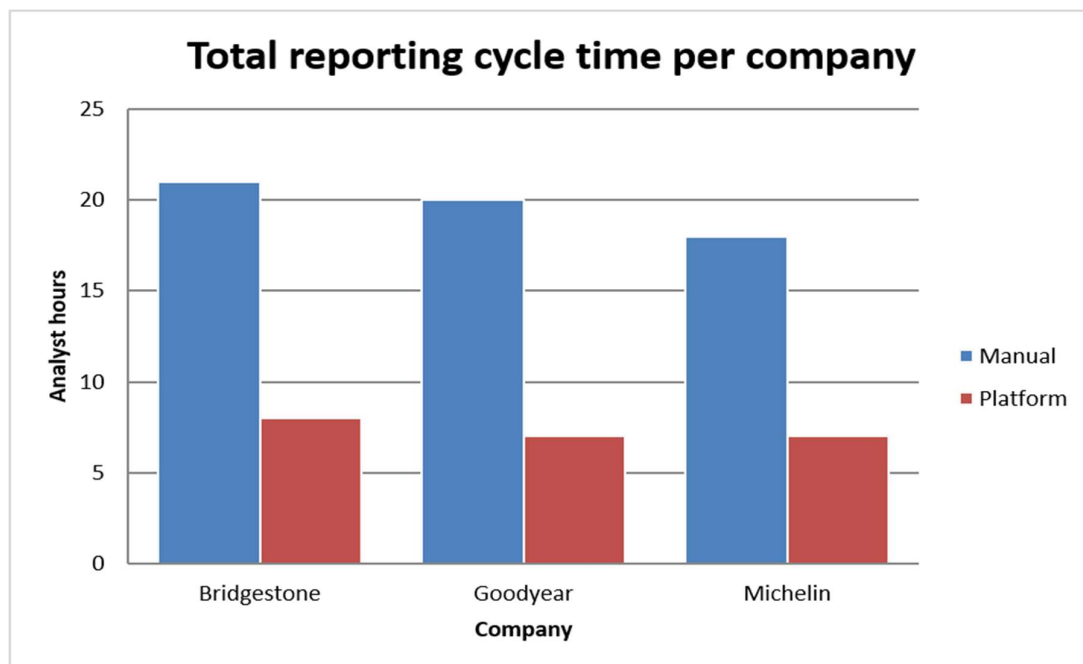


Figure 4.6 — Total reporting cycle time per company

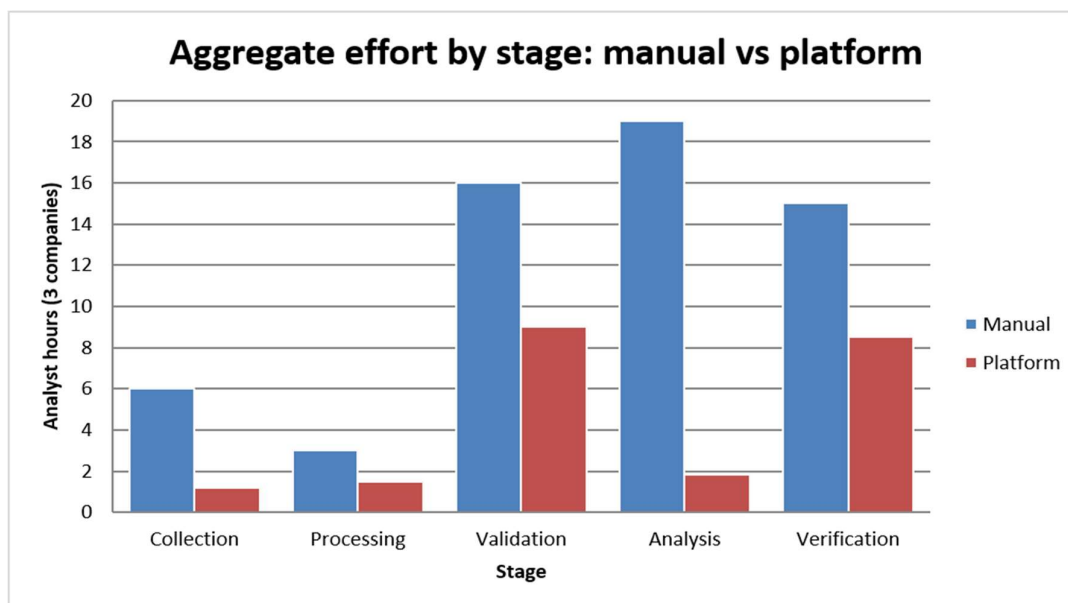


Figure 4.7 — Aggregate effort by stage: where the hours go

The productivity implication follows directly. In the manual process, most of the roughly 200 hours a full ten company manual cycle consumed were absorbed by collecting, converting, reconciling and checking data, which is preparatory work rather than analytical work. Under the platform that preparatory burden falls to roughly fifteen hours, and the analytical outputs that formerly had to be constructed by hand, namely the benchmarking packs and the sector report, are available continuously and at no marginal cost in time. The effect is not merely that reporting became faster. It is that the ratio between preparation and analysis is inverted.

4.6 Analytical Capability Enabled

The consolidated dataset supports analysis that the manual process could not sustain at reasonable cost.

At sector level, the ten companies in 2025 consumed 304.0 million GJ of energy and emitted 19.69 million tonnes of CO₂, of which 6.67 million tonnes were Scope 1 and 13.02 million tonnes Scope 2, against production of 26.48 million tonnes and water intake of 227.7 million cubic metres. Sector CO₂ intensity, computed as total emissions divided by total production, stands at 0.744, energy intensity is 11.48 and water intensity 8.60 (Table 4.7). Renewable sources account for 41.2 per cent of purchased electricity. Waste recovery is omitted from the sector position, since the metric is populated for only one company in one year and cannot support an aggregate.

Metrics	Value
Total energy	304.0 million GJ
Total emissions	19.69 Mt CO ₂
Scope 1 Emissions	6.67 Mt
Scope 2 Emissions	13.02 Mt
Production	26.48 million tonnes
Water intake	227.7 million m ³
CO ₂ intensity	0.744 tCO ₂ per tonne
Energy intensity	11.48 GJ per tonne
Water intensity	8.60 m ³ per tonne
Renewable share of purchased electricity	41.2%

Table 4.7: Sector position, 2025

Across the full series, sector CO₂ intensity has fallen from 0.871 in 2009 to 0.744 in 2025, a decline of 14.6% (Figure 4.8), with the steepest single year improvement occurring in the most recent period.

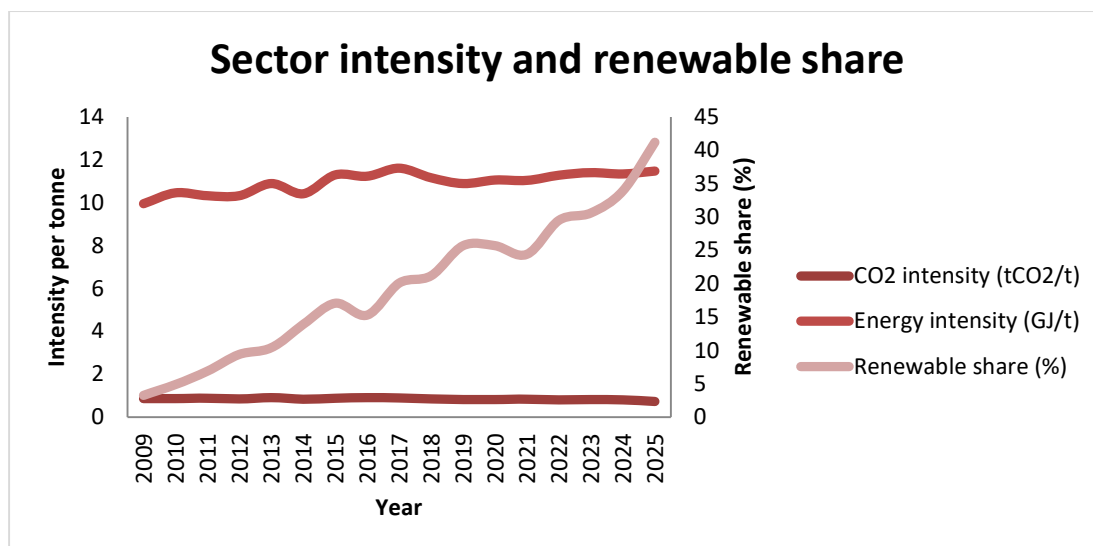


Figure 4.8 - Sector intensity and renewable share, 2009 to 2025

At company level, the dataset reveals a substantial dispersion in performance that the sector average conceals. In 2025, CO₂ intensity ranged from 0.604 at the best performing member to 1.015 at the weakest, a spread of 1.68 times (Table 4.8, Figure 4.9), with a median of 0.767 and an interquartile range from 0.619 to 0.823. This dispersion is precisely the information a benchmarking consortium exists to surface, and it is generated by the platform as a standing view rather than as a periodic analytical exercise.

Company	CO ₂ intensity	Energy intensity	Water intensity	Renewable %
Michelin	0.604	10.08	10.61	48.6
Apex Tire Co	0.606	7.56	8.35	20.9
Bridgestone	0.614	9.53	8.02	35.4
Summit Polymer Industries	0.635	9.97	6.17	44.5
Meridian Rubber Group	0.719	10.01	8.73	30.7
NorthStar Tyres	0.815	12.76	5.96	45.9
TerraGrip Industries	0.819	12.99	10.21	42.6
Velocity Rubber Corp	0.824	14.25	7.31	50.2
Pinnacle Elastomers	0.943	13.55	6.83	30.7
Goodyear	1.015	15.22	11.17	43.0
Sector	0.744	11.48	8.60	41.2

Table 4.8 - Company performance, 2025

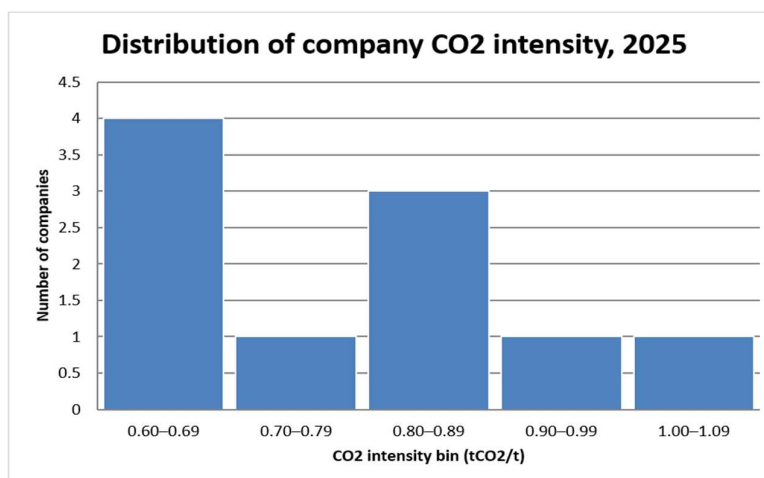


Figure 4.9 - Distribution of company CO₂ intensity, 2025

5. Discussion

5.1 Interpreting the findings

Three findings carry the weight of this study.

The first is that the efficiency gain is large and is attributable to redesign rather than to speed. The measured reduction of between 61 and 65% across the three companies of the parallel run is not the product of a faster spreadsheet. It arises because three activities that formerly consumed analyst time, namely unit conversion, KPI calculation and consolidation, no longer occupy any analyst time at all. They were not accelerated but relocated into software and executed on save. This distinction is the difference between a transactional and a transformational use of technology.

The second is that accuracy and efficiency did not trade against one another. A common and reasonable suspicion of automation is that speed is purchased at the cost of care. The evidence here points the other way. The same architectural decision that eliminated consolidation labour, namely centralising conversion and calculation in a single engine, is what made a 100% conversion pass rate attainable, because a rule applied once in code cannot be applied inconsistently across ten spreadsheets.

The third, and the most interesting, is that the platform's value lies substantially in what it makes visible. The 97 anomalies, the 1.68 times dispersion in company intensity, and above all the unit misnaming errors, were all present in the underlying data before the platform existed. They were simply invisible. The platform did not create this information. It made it discoverable.

5.2 The findings against literature

The results engage directly with the literature reviewed in Chapter 2. They substantiate Hammer's (1990) contention that the returns to automation derive from redesigning a process rather than accelerating it, since the elimination of consolidation labour followed from removing the stage entirely, not from performing it more quickly. In the terms of Davenport and Short (1990), the platform is transformational rather than transactional, in that continuously available benchmarking and sector aggregates constitute capabilities the manual process did not possess rather than faster versions of things it did.

The unit misnaming episode of Section 4.4.2 bears directly on Wang and Strong's (2015) argument that data quality is multidimensional. The affected records were complete, timely and internally consistent yet inaccurate, an error detectable only by a framework that separates accuracy from consistency and a system holding an independent expectation of the correct value. The unit heterogeneity of Section 4.2 is, in turn, the intra consortium counterpart of the measurement divergence that Berg, Kölbel and Rigobon (2022) found to account for the majority of ESG rating disagreement. The platform cannot resolve that divergence across an

industry, but within the consortium it dissolves it by imposing a common basis before any comparison is drawn.

The governance findings uphold rather than contradict the cautions of Kokina and Blanchette (2019) and Schlegel et al. (2024) that automation yields its benefits only where oversight adapts, since flagged anomalies are not resolved by the platform but routed to an analyst through the verification queue and recorded in an immutable snapshot. The study further supports Nezami et al. (2022), whose barriers to inter organisational collaboration, namely weak data sharing governance and poor standardisation, are precisely what the enforced schema and audit trail address, and Christensen, Hail and Leuz (2021), whose finding that standardisation improves reliability only when coupled with enforcement is instantiated by a conversion engine that imposes rather than recommends a common basis. In doing so the study supplies what Pizzi et al. (2023) and Münch et al. (2025) could not, the former having measured perceptions of accuracy without a spreadsheet baseline and the latter having derived design principles without validating them, whereas the present work implements such a platform and measures it against the process it displaced.

5.3 Recommendations

The recommendations proceed from specific to the general, addressing first the organisation that commissioned the platform and then any consortium contemplating a comparable system.

For dss+ and the Tire Industry Project, two measures follow immediately from the findings. First, unit declaration should be mandatory at the point of entry, with an absent unit treated as a submission error rather than an assumption. The 470 records currently lacking a recorded unit are handled by assuming they are already expressed in common units, a defensible convention for historical backfill but an undetectable source of error if extended to new submissions, and one closed at negligible cost by requiring the field. Second, the freed analyst capacity should be redirected deliberately rather than allowed to dissipate. The platform releases a substantial share of each reporting cycle from preparatory work, and that capacity yields value only if directed toward the interpretive questions the consortium exists to answer, such as the pronounced dispersion in carbon intensity between its strongest and weakest members.

For any organisation implementing an automated reporting platform of this kind, two further recommendations generalise beyond the present case. A ground truth fixture should be retained as a permanent regression test, since the assurance that conversions are correct depends on the existence of known correct answers, which real operational data does not provide; preserving a synthetic fixture that must pass before any change to the conversion logic is deployed sustains that guarantee indefinitely. And the convention by which any aggregate figure is computed should be defined and documented explicitly, since a sector level metric may be derived either as an unweighted mean of members or as a production weighted ratio, and the two diverge materially. A consortium that publishes a headline figure without stating which basis it uses invites misinterpretation, the

more so because an unweighted mean permits a small member to influence the reported average as heavily as a large one.

5.4 Feasibility, cost & implementation

A notable feature of this study is that the platform was designed, built, tested and deployed by a single analyst using Python, Streamlit and Microsoft Azure. Since the literature reviewed in Chapter 2 concentrates on large enterprises with mature technology functions, it might be inferred that systems of this kind demand comparable resources; the present case suggests otherwise, at least at consortium scale.

The costs are of three kinds. Infrastructure, comprising Azure App Services, SQL and Blob Storage, sits within the lower consumption tiers for a dataset of this size. Development was the principal cost and was borne once. Maintenance is the recurring one, and it is here that the honest assessment lies, since the platform creates a dependency on expertise in the conversion factors, emission factor tables and calculation logic. This dependency is mitigated but not removed by an extraction routine that regenerates the factor tables from the WBCSD source workbook rather than requiring manual transcription. Set against the analyst time released each cycle the arithmetic is favourable, but the saving is realised only where governance adapts, and Schlegel et al.'s (2024) finding that the greater part of automation success is organisational rather than technical is better read as a caution than a reassurance.

5.5 Limitations & critical reflection

The value of a critical evaluation lies less in cataloguing weaknesses than in showing how the design anticipates the problems it cannot fully escape. The limitations below are therefore presented as tensions the study is aware of, together with the manner in which the platform's design responds to each and the residual caveat that remains.

The first concerns the basis of the efficiency claim. Reporting effort is notoriously difficult to measure retrospectively, and a naive before and after comparison across different companies would confound the platform's effect with the idiosyncrasies of each firm. The study responds to this with a controlled parallel run, in which three companies performed the same task under both methods, isolating the platform's contribution. The residual limit is one of scale, since three of ten companies is a robust illustration rather than a generalisable measurement, and the figures for the remaining seven are analysts' own recollections. That the researcher was also the developer, and that the participating companies were already familiar with the platform, are sources of bias the design does not neutralise and that a larger independent study would.

The second concerns accuracy. The difficulty with any accuracy claim is the absence of a known correct answer against which to test. The study responds by constructing a synthetic dataset carrying ground truth values, against which the platform passed every one of 3,910 conversions. The residual caveat is that this

establishes correctness given a correctly recorded unit, and the misnaming episode of Section 4.4.2 shows that companies do not always record units correctly. The platform's guarantee is conditional, and the condition is human.

The third concerns the data itself. Precisely because ground truth can only exist for data whose answers are known in advance, the accuracy evaluation runs on synthetic rather than real submissions, which are messier. The design responds by subjecting the migration to real client data to the same validation and quality gates, which it passed, though this is assurance rather than proof. More broadly, the study observes one platform, in one industry, at one point in time, and cannot speak to whether the efficiency gain persists once novelty fades.

These limits are sharpened by the developmental character of the work. As a Design Science project, the platform matured across successive iterations, from an initial build on synthetic data, through refinement of the schema and conversion logic, to migration onto real client data and deployment on Azure. This dissertation evaluates and reports the final iteration only. The earlier iterations, though integral to how the artefact reached its assessed state, are not evaluated here, and the study therefore describes a system at a particular and relatively early moment in its life rather than one observed over time.

The most fundamental limit is one no design can overcome. The platform automates the production of ESG data but does not verify that the data is true. A company that misreports its emissions will have those figures converted, validated for internal consistency, consolidated and benchmarked with complete precision. Automation improves the fidelity of the pipeline; it does not confer honesty on the inputs, and no system of this kind should be represented as though it does.

5.6 Directions for further research

Three lines follow naturally. A larger, independently conducted timing study across all member companies would convert the remaining estimates into measurements. A longitudinal study across several reporting cycles would establish whether the gains persist and whether the freed capacity is genuinely redirected. And the AI assistant, once mature, invites its own evaluation, particularly of whether analysts can distinguish a reliable machine generated insight from a plausible but unfounded one.

Technical Appendix A: Supporting Detail for Chapter 1 & 2

This appendix expands details summarised in Chapters 1 and 2. It is provided for completeness and is not part of the assessed word count. Appendix B contains the platform implementation detail supporting Chapter 3.

A.1 The ESG reporting standards landscape

Framework	Body	Focus	Status
GRI	Global Reporting Initiative	Multi stakeholder impact disclosure	Voluntary, widely adopted
SASB	Sustainability Accounting Standards Board	Financially material, industry specific metrics	Voluntary, investor oriented
TCFD	Task Force on Climate Related Financial Disclosures	Climate risk governance and strategy	Voluntary, increasingly mandated
ISSB (IFRS S1, S2)	International Sustainability Standards Board	Global baseline for sustainability disclosure	Emerging, consolidating SASB and TCFD
CSRD	European Union	Mandatory detailed disclosure	Mandatory within the EU

Table A.1 - The ESG reporting standards landscape

A.2 The Wang and Strong data quality framework in full

Wang and Strong (2015) derived a hierarchical construct through a two-phase sorting study with data consumers. The four categories and their constituent dimensions are as follows.

Intrinsic quality, concerning the data, comprises accuracy, objectivity, believability and reputation.

Contextual quality, concerning the data in relation to the task, comprises relevance, timeliness, completeness, and the appropriate amount of data.

Representational quality, concerning the presentation of the data, comprises interpretability, ease of understanding, representational consistency and concise representation.

Accessibility quality, concerning the retrieval of the data, comprises accessibility and access security.

Chapter 3 operationalises four of these dimensions, namely accuracy, completeness, consistency and timeliness, on the grounds that they are the dimensions on which an automated platform can be expected to act directly and on which a comparison against a manual baseline is meaningful.

A.3 Wang's Total Data Quality Management cycle

Wang (1998) proposed a four-stage cycle, adapted from the continuous improvement tradition. The define stage identifies the data product and its quality requirements. The measure stage establishes metrics against those requirements. The analyse stage identifies root causes of quality deficiency. The improve stage implements corrective change, after which the cycle repeats. The platform evaluated in this dissertation embeds the measure and analyse stages directly in software, through its validation rules and readiness scoring.

A.4 Decomposition of ESG rating divergence

Berg, Kölbel and Rigobon (2022) analysed six major rating agencies, namely KLD, Sustainalytics, Moody's ESG, S&P Global, Refinitiv and MSCI.

Source of divergence	Contribution
Measurement methodology	56 per cent
Scope definition	38 per cent
Weighting	6 per cent

Table A.4 - Decomposition of ESG rating divergence

Pairwise correlation between raters ranged from 0.38 to 0.71. The dominance of measurement divergence indicates that the problem originates in how a firm's impact is measured rather than in how the resulting scores are combined, which is precisely the class of problem that a platform enforcing common definitions and unit conversion can address within a consortium.

Table A.5 — Complete synthesis of the reviewed literature

Sources are ordered chronologically, which lets the reader see the field develop from process reengineering, through data quality theory, to the recent and very thin evidence on ESG platforms themselves.

#	Study	Theme	Focus and Key Findings	Methodology	Limitations and Gap Addressed
1	Davenport and Short (1990)	Process automation	Distinguishes transactional, informational,	Comparative case analysis	Pre-dates modern automation.

#	Study	Theme	Focus and Key Findings	Methodology	Limitations and Gap Addressed
			analytical and transformational roles for information technology. Largest gains accrue from transformational use.	across nineteen companies	Adopted here as the primary evaluative lens for separating redesign from mere acceleration.
2	Hammer (1990)	Process automation	Radical redesign cut Ford's accounts payable staff from over 500 to 125, a 75 per cent reduction, by obliterating the process rather than mechanising it.	Conceptual argument built on the Ford case	Illustrative rather than measured. Supplies this study's central question, whether the platform redesigns work or merely relocates it.
3	Wang and Strong (2015)	Data quality	Establishes the canonical data quality construct across intrinsic, contextual, representational and accessibility categories.	Two phase sorting study with data consumers	General purpose, and silent on ESG specific problems such as scope boundaries and unit conversion. Operationalised here across accuracy, completeness, consistency and timeliness.
4	Elkington (1997)	Foundations	Introduces the triple bottom line, establishing environmental and social performance as legitimate objects of corporate reporting.	Conceptual monograph	Foundational. Provides the rationale for ESG reporting but no guidance on its operational execution.

#	Study	Theme	Focus and Key Findings	Methodology	Limitations and Gap Addressed
5	Wang (1998)	Data quality	Proposes the Total Data Quality Management cycle of define, measure, analyse and improve.	Conceptual framework, product perspective on data	Process oriented and generic. The platform embeds the measure and analyse stages directly in software.
6	English (1999)	Operational burden	Manual workflows generate misplaced decimals, inconsistent conversions, duplicate records and incomplete validation.	Practitioner monograph, information quality theory	Pre-dates ESG as a discipline and is general to data warehousing. Applied here to ESG unit conversion and consolidation.
7	Veleva and Ellenbecker (2001)	Consortia	Proposes a multi-level sustainable production indicator framework enabling comparison across firms.	Conceptual framework development	Prescriptive, with no evidence that firms adopt the indicators consistently. Reinforces the case for a platform that enforces rather than recommends definitions.
8	Veleva, Hart, Greiner and Crumbley (2003)	Consortia	Of six pharmaceutical firms, most reported only eco efficiency indicators, rarely extending to life cycle or carrying capacity measures.	Benchmarking study using a five level indicator hierarchy	Small sample, single industry and dated. Demonstrates that voluntary harmonisation of measurement is rarely achieved.

#	Study	Theme	Focus and Key Findings	Methodology	Limitations and Gap Addressed
9	Amoako-Gyampah and Salam (2004)	Digital transformation	Extends the Technology Acceptance Model to ERP, showing that perceived usefulness and shared belief drive adoption.	Survey based structural equation modelling	Concerns user acceptance rather than measurable performance. Informs the credibility dimension.
10	Hevner, March, Park and Ram (2004)	Methodology	Establishes seven guidelines for design science research producing and evaluating purposeful artefacts.	Conceptual framework, information systems paradigm	Supplies the overarching research strategy.
11	Peppers, Tuunanen, Rothenberger and Chatterjee (2014)	Methodology	Operationalises design science as six sequential activities, from problem identification to communication.	Process model synthesis, demonstrated across four case studies	Provides the procedural spine the project follows.
12	Seddon, Calvert and Yang (2010)	Digital transformation	Most factors determining whether enterprise systems deliver benefits are organisational rather than technical.	Multi project benefits realisation model, case based	Focused on large enterprise systems. Reinforces the large organisation bias this study addresses.
13	Westerman, Bonnet and McAfee (2011)	Digital transformation	Gains follow from redesigning processes around digital capability rather than layering technology onto existing ones.	Multi case research across large enterprises	Concentrates on large firms with mature IT, leaving mid-market firms, consortia and consultancies under examined.

#	Study	Theme	Focus and Key Findings	Methodology	Limitations and Gap Addressed
14	Hahn and Kühnen (2013)	Operational burden	Maps sustainability reporting adoption onto legitimacy, stakeholder, signalling and institutional theory, and identifies a persistent gap between frameworks and implementation.	Systematic literature review of 178 articles, 1999 to 2011	Predates the shift to mandatory reporting. Concerns why firms report, not how reports are produced.
15	Fitzgerald, Kruschwitz, Bonnet and Welch (2014)	Digital transformation	Organisational rather than technical barriers determine whether digital adoption succeeds.	Survey and interview study of managers and executives	Broad in scope and not specific to reporting workflows.
16	Friede, Busch and Bassen (2015)	Foundations	ESG performance is materially linked to financial performance, which underpins investor scrutiny of disclosure credibility.	Meta analysis aggregating over 2,000 empirical studies	Concerns the outcomes of ESG performance, not the reliability of the underlying data. Motivates the credibility dimension.
17	Venable, Pries-Heje and Baskerville (2016)	Methodology	The FEDS framework separates artificial from naturalistic, and formative from summative, evaluation.	Framework development with illustrative applications	Justifies the dual evaluation design of controlled accuracy testing plus naturalistic before and after comparison.
18	DAMA International (2017)	Data governance	Establishes role-based access control, encryption, audit trails and approval workflows	Practitioner body of knowledge, consensus standard	Normative rather than empirical. Specifies what governance

#	Study	Theme	Focus and Key Findings	Methodology	Limitations and Gap Addressed
			as standard governance requirements.		requires without evaluating whether platforms deliver it.
19	Kokina and Blanchette (2019)	Process automation	RPA reduced errors, lowered costs and enforced standardisation, but only where accompanied by process redesign and new oversight structures.	Interview study of adopters, framed by Task Technology Fit and Technology to Performance Chain theory	Confined to accounting and finance, and qualitative. Supplies the governance caution central to the credibility dimension.
20	Christensen, Hail and Leuz (2021)	Disclosure quality	Mandates improve disclosure quantity and reach, but effects on comparability and reliability are mixed and context dependent.	Economic analysis and literature review across jurisdictions	Concerns market level regulation, not the systems producing disclosures. Establishes that standardisation without enforcement is insufficient.
21	Berg, Kölbel and Rigobon (2022)	Data quality	Measurement methodology drives 56 per cent of ESG rating divergence, scope 38 per cent and weighting 6 per cent. Pairwise correlations range from 0.38 to 0.71.	Quantitative decomposition mapping six rating methodologies onto a common taxonomy	Analyses divergence between external raters, not within a reporting organisation. Motivates enforcing common definitions and unit conversion within a consortium.

#	Study	Theme	Focus and Key Findings	Methodology	Limitations and Gap Addressed
22	Nezami, de Bruijne, Hertogh and Bakker (2022)	Consortia	Collaboration benefits are hampered by weak data sharing governance, information asymmetry, misaligned incentives and poor standardisation.	Systematic literature review with practitioner interviews	Set in infrastructure construction, not ESG. This study tests whether a platform enforcing validation and audit trails overcomes the same barriers.
23	Tawiah and Borgi (2022)	Disclosure quality	XBRL adoption improves information efficiency, comparability and display quality, most strongly in developing countries.	Global panel study of XBRL adoption in financial reporting	Concerns financial rather than ESG reporting. Provides the precedent this study extends.
24	World Economic Forum (2022)	Operational burden	ESG metrics originate in fragmented sources using differing definitions, protocols and timelines, with data historically collected by circulating spreadsheets, making consolidation laborious and error prone.	White paper, interview based case studies of ten reporting companies	Descriptive and normative. Diagnoses fragmentation without measuring its cost or testing a remedy.
25	Schlegel, Rosenberg, Fundanovic and Kraus (2024)	Process automation	Technical implementation accounts for only around 30 per cent of variance in	Synthesis of success factors across RPA and BPA projects	Concerns project success generally, not reporting quality. Justifies

#	Study	Theme	Focus and Key Findings	Methodology	Limitations and Gap Addressed
			automation project success. The remainder is organisational and governance related.		measuring changes in roles, workflows and governance alongside efficiency.
26	Pizzi, Mastroleo, Venturelli and Caputo (2023)	ESG platforms	Adoption of Sustainable ERP systems leads analysts to more accurate evaluations of environmental disclosure. Among the few peer-reviewed platform evaluations.	Fuzzy expert system grounded in contingency theory, Italian FTSE MIB firms	Measures perceived accuracy, not operational efficiency, and does not benchmark against spreadsheets. This dissertation supplies that direct comparison.
27	Münch, Mayer and Quick (2025)	ESG platforms	Derives ten design principles for a harmonised ESG reporting platform serving external compliance and internal management accounting.	Design Science Research across two implementation cycles, single organisation	Prescriptive rather than evaluative. No empirical validation that the principles deliver the promised performance. This dissertation implements and evaluates such a system.
28	Commercial platforms: Workiva, SAP, Plan A	ESG platforms	Deployed across hundreds of organisations on the strength of vendor performance claims.	Vendor documentation and practitioner deployment	No peer-reviewed evaluation exists, despite the considerable sums invested.

Table A.5 - Complete synthesis of the reviewed literature

Technical Appendix B — Platform Implementation Detail

This appendix expands the implementation detail summarised in Sections 3.3 and 3.4. It is provided for completeness and reproducibility and is not part of the assessed word count.

B.1 System architecture and the storage-abstraction layer

The platform is implemented in Python (Streamlit front end) and follows a layered architecture in which each concern is isolated behind a stable interface. Persistence is handled by a storage layer exposing a single command set (read, write, list, archive) that is identical across implementations as a local file store and a SharePoint document library during development, and Azure Blob Storage with Azure SQL in production. Documents flow through a fixed folder hierarchy 01_Templates_Raw → 02_Validated → 03_Consolidated → 04_Reports → 99_Archive that mirrors the reporting lifecycle. All data reads are routed through one shared data-access module, so that the change from file-based reads to Azure SQL queries is confined to that module and leaves the page-level code untouched. This separation of concerns is the principal reason a system of this size remained maintainable under single-developer conditions, and it is what made the local-to-cloud migration a configuration change rather than a rewrite.

B.2 Data schema: long-form, wide and the three-column unit structure

The canonical store is a long-form table with the columns Company, Year, KPI_Name, Value, Unit (approximately 224 KPI fields per company-year, ~38,080 rows across the ten companies). A canonical field registry maps each unit bearing field across its four names, namely the calculation attribute, the long form KPI_Name, the conversion table subsection and the wide table column, ensuring that a single field cannot drift out of alignment between the submission form, the engine and the display tables.

For each of the 23 unit-bearing fields the wide table stores three columns starting with the corporate value (<Field>_CorporateValue), the reported unit (<Field>_Unit), and the common value (<Field>, retained under its original name so existing readers are unaffected). Worked example: if a company enters electricity as 100 megawatt-hours, the platform stores 100 and MWh as the corporate value and unit, and 360 as the common value, since 1 MWh = 3.6 GJ. The fixed common units are gigajoules for energy, metric tonnes for mass-based metrics, and cubic metres for water. Non-unit fields (counts, percentages, pre-fixed figures such as CO₂ totals) pass through unchanged as a single column.

The synthetic fixture adds a GroundTruthCommonValue column giving the exact common-unit figure each conversion must reproduce; this column is absent from production data.

B.3 The data-processing pipeline

Conversion factors are extracted directly from the WBCSD KPI Collection Tool's own conversion sheets into a UNIT_CONVERSION_FACTORS table keyed by (subsection, unit), alongside an IEA grid-emission-factor taxonomy of roughly 150 country and region entries by year. On submission, the pipeline (i) converts every unit-bearing field to its common unit; (ii) computes derived KPIs (intensities, totals, shares); (iii) writes the long-form and wide tables together; and (iv) recomputes sector aggregates live. Imperfect data is handled explicitly, so that a field with no recorded units is treated as already in the common unit (a documented fallback for historical records), while country electricity columns that are zero for every company are dropped to avoid clutter.

Validation rules include waste recovery $\leq$ total waste, ISO-certified sites $\leq$ total sites, and a year-on-year movement flag triggered beyond a set threshold (20%). The ground-truth round-trip routine rebuilds the wide table from the long-form data, compares each converted value with its Ground Truth Common Value, and raises an error on any difference exceeding a rounding tolerance (relative 1×10^{-6}), causing the build to fail rather than ship a silent conversion error.

B.4 The KPI-calculation and emission-factor engine

The engine is a pure calculation module (no interface or file dependencies). Scope 1 emissions are computed fuel by fuel using IPCC emission factors (tonnes CO₂ per GJ, lower heating value), coal is tracked as three sub-types (sub-bituminous, brown-coal briquettes, other bituminous) and fuel oil as two (Heavy A, Heavy C), each carrying its own factor, with a legacy blended-total fallback retained for backward compatibility. Scope 2 is derived from non-renewable purchased electricity via the location-based IEA grid factor (a configurable default of 0.45 t CO₂/MWh, the European average, overridable per company). The engine returns Scope 1, Scope 2 and total CO₂, total energy, and the three intensity KPIs (energy, CO₂ and water per tonne of production), together with validation flags graded by severity (error, warning, ok).

B.5 Validation, readiness scoring, the save chain and the audit trail

The readiness score combines section-level completeness with the count and severity of validation flags into a single 0–100 figure with a categorical label. In the analyst verification queue each flag may be accepted or queried, and the submission approved once resolved. The save chain, on each submission, updates the wide and long-form masters, regenerates the consolidated workbook and per-company files, and writes an immutable timestamped snapshot of the company's full record. A parallel field-level change-comment workflow lets a company annotate overwritten data and an analyst mark it Pending, Seen, Rejected or Approved and every comment event is recorded, giving a complete, reviewable provenance trail for each figure.

B.6 The two-shell role model and page inventory

The client workspace comprises Home, My Dashboard, My Records, Benchmarks, Reports, Submit Data and Settings; the dss+ internal workspace comprises Portfolio, Company Data, Verification Queue, Analysis, Benchmarks, AI Assistant, Document Library, Sector Reports, Admin and Settings. Access is role-based, so a client reaches only its own records while a dss+ analyst holds cross-company read access and verification-write permissions.

Technical Appendix C: Generative AI Usage Statement

This statement documents the use of generative AI in the preparation of this dissertation, in the interest of full transparency. It is not part of the assessed word count. In every instance, AI was used in an assistive capacity. All research decisions, the design and construction of the platform, the analysis of results and the intellectual argument are the author's own. Every AI output was reviewed, verified and, where necessary, corrected by the author, who takes full responsibility for the final work.

C.1 Overview and declaration

Generative AI tools were used at several stages of this project as an aid to thinking, drafting and debugging. The principal tools were a large language model assistant, used for ideation, literature triage and improving the clarity of the author's own writing, and GitHub Copilot, used for isolated code suggestions and bug fixes during platform development. AI was not used to generate research findings, to make design or analytical decisions, or to author the substantive content of the dissertation. Its role was to accelerate routine work and to surface options that the author then evaluated, accepted, modified or rejected.

Two boundaries were maintained throughout. First, no confidential client data was entered into any external AI tool at any point. Development and testing used the synthetic dataset, and the platform's own AI assistant operates only on anonymised, derived summaries, as described in Section 3.4.3. Second, the platform backend and its core logic were written by the author, with AI assistance in development confined to suggesting fixes for specific, isolated errors.

The subsections below describe where and how AI was used in each phase. The consolidated log in Section C.6 documents representative interactions, giving the prompt, the output received and the change the author made. As that log shows, a recurring and important pattern was the identification and correction of flawed AI output, which is recorded as evidence of critical oversight exercised.

C.2 Ideation and scoping

At the outset the project could have taken several forms, including a forecasting model for ESG scores, a tool for extracting publicly available sustainability data, or an automated reporting platform. AI assistance was used to help articulate and weigh the trade-offs of each option, considering data availability, novelty and fit with the consulting context. The decision to pursue the reporting platform was the author's, taken based on its fit with the internship and the identified research gap. A forecasting model was set aside in part because the available sample, ten companies across seventeen years, is too small to support reliable forecasting. Where AI helped to frame the research question and objectives, the output was rewritten by the author and restructured so that the four objectives map cleanly onto the Design Science stages.

C.3 Literature review

AI assistance was used to identify candidate papers and to summarise a wider body of work than appears in the final review, which supported the author's judgements about relevance. Every cited source was located, read and verified against the original. This verification was not a formality. It caught citations that were inaccurate or misattributed, including one whose stated topic did not match the actual paper, and led the author to check every reference in the dissertation against its source. AI was used only to triage and shortlist; the decisions about which sources were pertinent, and the reading and interpretation of them, were the author's own.

C.4 Data architecture, visualisation and development

AI assistance served as a sounding board while the data architecture was settled, namely the choice to maintain both a long form production store and a wide analytical table. The final schema, including the three-column unit structure that stores the corporate value, its unit and the converted common value, was designed and implemented by the author. During interface and visualisation work, AI assistance was used while writing Plotly charting code and laying out Streamlit elements, with the author deciding what to present and how. The platform backend was built by the author. GitHub Copilot contributed only isolated bug fixes and function completions, each of which was tested within the existing architecture and, in the case of the unit conversion logic, checked against the WBCSD conversion tables and the ground truth test.

C.5 Write up

Across the methodology, results and discussion, AI assistance was used to improve the clarity and academic register of the author's own drafts, and to check that each finding in the discussion was linked back to a source in the literature review. Every suggestion was reviewed, and only those faithful to the intended meaning were retained. The arguments, findings and interpretations were made by the author. Specific focus was to ensure that AI assisted rephrasing did not overstate any finding, especially the efficiency results, where the manual figures are estimates for seven companies and measured for only three.

C.6 Consolidated usage log

The table below records representative interactions across each phase. For each, it gives the author's prompt, a summary of the AI output, and the author's decision, whether accepted, modified or rejected, with the reasoning. The pattern of modification and rejection, and in particular the correction of inaccurate citations and analytical output, is the evidence of the critical judgement and oversight that governed the use of these tools.

#	Phase	Prompt (author)	AI output	Decision and reasoning
I1	Ideation	Compare three possible ESG analytics projects, a forecasting model for sustainability scores, a public data extraction tool, and an automated reporting platform, with the pros and cons of each for a consulting internship context.	Outlined each option's data demands, novelty and practical fit.	Accepted as input. The final choice of the reporting platform was the author's, made against the identified research gap.
I2	Ideation	What is the research gap if the literature already covers ESG frameworks and automation separately?	Suggested the gap lies at their intersection, in measuring operational change when a real workflow is automated.	Modified. Reframed in the author's own words and sharpened to the spreadsheet baseline comparison that no prior study performs.
I3	Ideation	Would a forecasting model be viable given only ten companies and seventeen years of data?	Noted the sample is too small for reliable forecasting and that overfitting would be a serious risk.	Accepted. This reasoning contributed to setting the forecasting option aside.
I4	Ideation	Frame a single research question and four objectives for a build and evaluate ESG platform study.	Drafted a question and four candidate objectives.	Modified. Rewritten by the author and restructured to map onto the Design Science stages.
I5	Ideation	What would make this project appropriate for an MSc rather than a routine tool build?	Suggested emphasising the controlled evaluation and grounding each measure in the literature.	Accepted as guidance. Informed the decision to anchor every metric in a Chapter 2 construct.
L1	Literature	Suggest recent peer reviewed work evaluating digital ESG reporting platforms.	Listed candidate papers, some relevant and several not.	Modified. Located and read each; retained Pizzi et al. (2023) and Münch et al. (2025), discarded the rest as off topic.

#	Phase	Prompt (author)	AI output	Decision and reasoning
L2	Literature	Summarise the method and key finding of this paper. (applied to several papers)	Produced short summaries of method and findings.	Verified. Each summary checked against the original before use; several were imprecise and corrected.
L3	Literature	Provide a citation for a source on best practices in sustainability reporting.	Produced a citation attributed to an author that, on checking, did not match the actual paper's topic.	Rejected. The citation was inaccurate and removed. The episode prompted full manual verification of every reference in the dissertation.
L4	Literature	Which theoretical frameworks are standard for measuring data quality and process automation?	Named Wang and Strong's data quality framework and the reengineering literature.	Accepted and verified against the primary sources, which were then read in full and cited directly.
L5	Literature	Summarise a larger set of automation in accounting papers so I can judge which are relevant.	Summarised a broad set, most ultimately unused.	Accepted as triage. Used only to shortlist; Kokina and Blanchette (2019) and Schlegel et al. (2024) were selected and read in full.
L6	Literature	Is there a management study specifically of the WBCSD Tire Industry Project's data process?	Indicated none appears to exist and that TIP's academic output concerns tyre road wear particles.	Verified independently by the author's own searching and stated as such in Section 2.7.
D1	Architecture	Discuss the advantages of storing reporting data in long form versus wide form for this use case.	Explained the scaling and analysis tradeoffs of each layout.	Accepted as input. The decision to retain both layouts was the author's.
D2	Architecture	How should I store a value that a company reports in one unit but the platform compares in another?	Suggested storing the original and converted values separately.	Modified. Developed into the three-column structure (corporate value, unit, common value) and implemented by the author.
D3	Visualisation	Write Plotly code for a grouped bar chart comparing a company's intensity against the sector average.	Provided boilerplate charting code.	Modified. Adapted to the platform's data structures and restyled; the author chose which comparisons to show.

#	Phase	Prompt (author)	AI output	Decision and reasoning
D4	Development	This Streamlit callback throws a state error on save; what might be wrong?	Suggested likely causes in the callback ordering.	Fixed. Applied and tested within the author's architecture; the design was unchanged.
D5	Interface	Suggest a layout for a submission form with per field unit selection.	Proposed a candidate arrangement of form elements.	Modified. Reworked to fit the WBCSD schema and the platform's two shell role model.
D6	Development	Copilot: complete this function that converts a value using a factor looked up by field and unit.	Produced a first pass conversion function.	Modified and verified. Rewritten against the WBCSD conversion tables and checked with the ground truth test.
W1	Write up	Improve the academic clarity of this paragraph without changing its meaning. (applied throughout)	Returned revised phrasings of the author's own drafts.	Modified. Only revisions faithful to the intended meaning were kept; the rest were rejected.
W2	Write up	Generate word count for each section.	Returned an approximate word count for each section.	Accepted as a rough estimate, because the approximate counts were already known.
W3	Write up	Suggest a clearer structure for the results chapter.	Proposed an ordering of sections.	Modified. Adapted to the author's own results and the objective by objective logic.
W4	Write up	Rephrase this claim so it does not overstate the finding.	Returned a more measured phrasing.	Accepted. Consistent with the author's intent to avoid overclaiming, especially on the manual estimates.
W5	Write up	Check this discussion section links each finding back to a Chapter 2 source.	Flagged two findings not yet tied to the literature.	Accepted as a prompt to revise. The author added the links, verifying each against the cited source.

Table C.1 - Consolidated generative AI usage log

C.7 Boundaries and responsibility

Generative AI was not used to design platform architecture, to make methodological or analytical decisions, to interpret results, or to generate or alter any data. The synthetic dataset was constructed by the author to match the WBCSD schema, and no real client data was exposed to any external AI tool. The final judgements, conclusions and recommendations in this dissertation are the author's own.

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